

Intrinsic finite elements for PDEs and spectra on Riemannian manifolds

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Abstract

We construct high-order finite elements for partial differential equations and spectra on compact Riemannian manifolds that are described by intrinsic data on a simplicial mesh: the metric tensor on each cell and the maps that pair cell facets. The unknown is written in broken Bernstein–Bézier form, conformity is a sparse linear constraint on local coefficients, and a sparse extension matrix reduces the constrained problem to a symmetric finite element system. The same data assemble compatible polynomial spaces of differential forms on quotient meshes, including cells glued to themselves; the discrete exterior derivative is independent of the metric and the Hodge matrices carry it. We prove that the constrained spaces form a complex, carry bounded commuting projections to finite free isometric quotients by averaging, and show that on a fixed complex a change of metric moves every positive discrete Hodge eigenvalue by explicit factors that are independent of the dimension of the space and of the polynomial degree; the bounds admit cellwise tensors with jumps. Scalar and one-form manufactured problems on a variable-metric three-torus and on a torus carrying the metric of a trefoil tube surface converge at the expected orders, through degree six for scalars and order four for one-forms. On five cells of the Poincaré dodecahedral space, degree fourteen gives the first eigenvalue with absolute error $3 \cdot 10^{-11}$ and the correct multiplicities of five levels. On the Seifert–Weber hyperbolic space the lowest coexact one-form eigenvalue is computed as 2.03927 with multiplicity six, inside the interval indicated by Lin and Lipnowski.

Keywords: intrinsic finite elements, Riemannian metrics, Bernstein polynomials, finite element exterior calculus, Hodge Laplacian, spectral approximation, high-order methods

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1. Introduction

Finite element approximation on a Riemannian manifold begins with the same questions as finite element approximation on a Euclidean domain: how to represent local trial functions, how to join them across elements, and how to assemble the weak problem. The main difference is the geometric information needed for integration. A Riemannian metric specifies lengths, volume, and inner products of derivatives in local coordinates. Many compact manifolds have no useful global coordinate system or prescribed Euclidean embedding and are presented through local charts or through identified faces of a mesh. A finite element method for them should take the metric and the identifications as its data.

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We use ordinary reference simplices throughout. A mesh supplies an elementwise coordinate map, the metric expressed in the element coordinates, and the correspondence between paired facets. The metric determines the element integrals, and the facet correspondences determine the assembly. Keeping these two inputs separate allows the same reference-element basis, sparse assembly routines, and differential operators to be reused when the metric changes, when a mesh is expressed in different coordinates, or when a domain is assembled using nonstandard facet identifications. The metric can be smooth, nonconformal, and anisotropic. The pullbacks of the facet maps must preserve the chosen finite-dimensional trace spaces. For the polynomial spaces used here this requirement is met by compatible affine reference-facet identifications.

Bernstein–Bézier polynomials give an especially transparent coefficient representation for this construction. Each element has its own polynomial coefficients. Continuity is imposed by relations between the coefficients on paired traces, and the independent global unknowns are obtained from those relations. This viewpoint follows the spline-theoretic treatment of smoothness and determining sets in Lai and Schumaker [29] and the constrained B-form method of Awanou, Lai, and Wenston [7]. The earlier conformal spherical spline method [27] uses the same principle to discretize the Laplace–Beltrami problem on genus-zero surfaces. The accompanying work on Bernstein constraint complexes [25] develops explicit edge smoothness functionals, their commuting relations with differentiation, and constrained Galerkin realizations for scalar and vector spline spaces on triangulated surfaces. Here the coefficient construction is used with intrinsic element metrics in arbitrary dimension. The local polynomials are ordinary reference-simplex Bernstein polynomials, and the assembly admits paired facets that need not be adjacent in a single coordinate display.

Scalar elliptic problems require trial functions that agree in value across a paired facet, and their element integrals use the metric to measure gradients and volume. Systems involving gradient, curl, and divergence require continuity of tangential or normal traces, which compatible polynomial differential-form spaces impose in a unified way. The exterior derivative commutes with pullback, so the reference-element trace relations assemble into a global differential complex whose inner products and adjoints are computed from the metric. We use the established local spaces and approximation theory of finite element exterior calculus [3–6, 28], with the high-order Bernstein implementations of Ainsworth, Andriamaro, and Davydov and of Ainsworth and Fu [1, 2]. The coefficient-level viewpoint of [25] supplies a complementary construction of the global constraints.

Prior work already treats intrinsic manifold FEEC and the transformation of high-order finite element degrees of freedom. In particular, the intrinsic commuting-projection theory of Licht [34, 35] provides the abstract form approximation framework; Gawlik and McKee [23] analyze the Hodge–Laplace source problem when the smooth metric is replaced by a Regge metric on the same manifold, with trace and inverse inequalities and mesh regularity stated through intrinsic metric comparison constants, and they bound the geometric consistency term by the L^∞ difference of the two metrics; the sub-entity permutation and degree-of-freedom transformations of Scroggs, Dokken, Richardson and Wells [37] are directly related to the matrices used for our form traces. Symmetric bases and their limits are studied by Licht and Berchenko-Kogan [12, 36]. For the Poincaré dodecahedral space, Bachelot-Motet [9] previously constructed group-invariant finite element spaces and computed scalar Laplace–Beltrami eigenvalues. Our numerical use of that manifold is a comparison of elementwise metric data, facet-constrained assembly, and compatible complexes on a small quotient mesh.

The local polynomial spaces, the Whitney maps, the abstract Hodge theory, and the commuting

projections on intrinsic triangulations are those of finite element exterior calculus [3–5, 34, 35], and intrinsic assembly from element metrics is established for surfaces and for Regge metrics [15, 20, 22, 23]. The present paper works inside that framework and contributes three things. The first is a coefficient-level realization: conformity is the linear constraint $J_k c = 0$ on broken Bernstein or moment coefficients, the sparse extension matrix Z_k is built from entity classes, quotient facets, and self-paired cells, it is reused when the metric changes, and the assembled exterior derivative satisfies the complex identities (Theorem 4.1, Section 4.2). The second is a set of fixed-space perturbation bounds: on a fixed finite-dimensional complex, a replacement of the inner products changes every positive discrete Hodge eigenvalue by factors that are independent of the dimension of the space and of the polynomial degree; the argument is purely algebraic and covers cellwise tensors with jumps (Theorems 5.9, 5.10, and 5.14). The third is a transfer of commuting projections to finite free isometric quotients by averaging (Theorem 4.6), which brings the quotient meshes of Section 8 into the scope of the cited projection theory; other mesh families enter through Assumption 5.5.

The predecessor on conformal spherical splines develops the scalar weighted weak form in dimension two [27], and its continuation to compact two-manifolds of positive genus treats the side pairings of a fundamental polygon as additional rows of the smoothness matrix [26]; the companion manuscript develops constraint complexes on triangulated surfaces [25]. The present work treats general intrinsic metric tensors, higher-dimensional facet identifications, the induced metric on polynomial differential forms, and comparisons of two metrics on the same finite element space. The trimmed polynomial spaces, Whitney maps, commuting projections, and the abstract source and spectral approximation theorems come from established finite element theory [3–5, 18, 24, 34], and Theorems 5.1 and 5.8 state their consequences under interpolation, regularity, and projection hypotheses. Theorem 4.6 supplies a commuting projection on a finite free isometric quotient from one on a compatible covering mesh.

The analysis has three parts: a conformity result that relates reference-facet coefficient constraints to global finite element functions and, through the pullback identity for the exterior derivative, to the compatible spaces; approximation and spectral convergence under mesh regularity and uniformly bounded commuting projections, which we transfer from compatible intrinsic triangulations and from covering meshes of finite isometric quotients; and fixed-space metric-perturbation bounds. These arguments draw on the finite element analysis of elliptic and eigenvalue problems [8, 13, 17], intrinsic complexes and commuting projections [18, 19, 34, 35], and geometric perturbation estimates [15, 20, 22–24].

Replacing a smoothly varying metric by a simpler approximation changes the bilinear forms even when the mesh, the approximation space, and the constraints are unchanged. We compare the two sets of forms and their discrete spectra directly, which separates geometric data error from polynomial approximation and numerical integration. Intrinsic and coordinate-free implementations of exterior calculus provide further context [11, 38].

The numerical experiments begin in two dimensions with a torus that carries the induced metric of a trefoil tube surface. Its non-diagonal metric is used in a manufactured Poisson problem, with convergence through degree six, and in Laplace–Beltrami eigenvalue computations through index 480. The three-dimensional experiments address accuracy for smooth variable metrics, the form complex, sensitivity to metric replacement, and quotient meshes. Manufactured scalar and one-form problems on a three-torus with nonconstant metrics measure convergence in the mesh size and the degree, and matched-mesh calculations compare the prescribed metric with a cellwise approximation. The Poincaré dodecahedral space tests facet identification, harmonic dimensions,

and spectral approximation on a five-cell mesh, and the Seifert–Weber hyperbolic space contrasts the model metric with a piecewise-flat replacement; there the lowest coexact one-form eigenvalue is computed as 2.03927 with six members, a numerical approximation consistent with the interval indicated by Lin and Lipnowski [31]. Solid tubes around a circle and a trefoil knot give scalar Dirichlet and Neumann problems. Figures of computed eigenfunctions accompany each manifold; on the two dodecahedral spaces the eigenfunctions are adapted to the rotation symmetries of the fundamental cell. The finite element formulation is given in dimension n , with numerical examples in dimensions two and three. Throughout, p denotes the degree of the full scalar Bernstein space $S_p^0(\Delta)$; r denotes the order of the trimmed differential-form spaces $V_{h,r}^k$.

The following sections develop the finite element data, the scalar and compatible spaces, approximation and spectral analysis, and the numerical results in that order.

2. Finite element setting

Write $\hat{K} = \{\hat{x} \in \mathbb{R}^n : \hat{x}_i \geq 0, \sum_i \hat{x}_i \leq 1\}$ and denote its barycentric coordinates by $b_0 = 1 - \sum_i \hat{x}_i$ and $b_i = \hat{x}_i$ for $i \geq 1$. We use *intrinsic element maps* $F_K : \hat{K} \rightarrow M$, smooth and nonsingular up to each cell boundary in the usual compatible piecewise-smooth sense. The pullback metric $g_K = F_K^*g$ is represented in reference coordinates by the matrix

$$G_K(\hat{x})_{ij} = g_{F_K(\hat{x})}((F_K)_*\partial_i, (F_K)_*\partial_j), \quad \rho_K(\hat{x}) = \sqrt{\det G_K(\hat{x})}. \quad (2.1)$$

Here $g = g_{ij}(x) dx^i \otimes dx^j$ in a chart, with the summation convention, and $F_K^*(dV_g) = \rho_K d\hat{x}$. The element routines use only G_K and the facet maps; an explicit chart of a quotient or of a geometric model is one way of providing G_K .

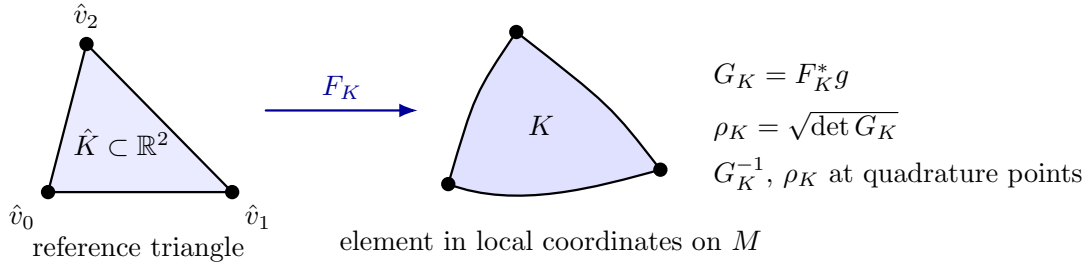


Figure 1: Reference simplex, intrinsic element map, and local metric data in dimension two. The element K is displayed in local coordinates on M .

Figure 1 shows these data.

Assumption 2.1 (Finite element metric and facet data). For a shape-regular mesh family there are constants $0 < \lambda_* \leq \lambda^* < \infty$, independent of K and h , such that in reference coordinates

$$\lambda_* h_K^2 |\xi|^2 \leq \xi^T G_K(\hat{x}) \xi \leq \lambda^* h_K^2 |\xi|^2 \quad (\xi \in \mathbb{R}^n),$$

with the usual equivalent anisotropic scaling by an element reference metric when the mesh family is not quasi-uniform. In particular G_K is measurable, symmetric, and positive definite. On each pair of identified facets $F = K^+ \cap K^-$, the maps $F_{K^\pm}$ induce a compatible homeomorphism $\tau_F : \hat{F}^+ \rightarrow \hat{F}^-$ such that

$$F_{K^+}|_{\hat{F}^+} = F_{K^-}|_{\hat{F}^-} \circ \tau_F. \quad (2.2)$$

The metric traces satisfy the resulting tensor change-of-coordinates rule. For high-order polynomial form assembly, τ_F is further assumed to preserve the chosen finite-dimensional trace spaces under pullback. In the implementation this is achieved by affine reference-facet maps induced by vertex permutations; the scalar C^0 construction admits more general compatible transition rules if its trace spaces are accordingly chosen.

The element tensors that occur in this paper fall into three classes. A *smooth* metric is the pullback of one Riemannian metric g on M and satisfies the tensor change-of-coordinates rule on every paired facet. A *Regge* metric is smooth on each cell and has single-valued tangential-tangential components on every shared facet, which is the condition (2.6) imposed facet by facet; its normal components may jump. This is the class for which Gawlik and McKee [23] develop L^2 and $H(d)$ spaces, intrinsic trace and inverse inequalities, and the Hodge–Laplace source error analysis. A *measurable cellwise* tensor is uniformly positive definite on each cell and is free across facets. The implementation measures the discrepancy in (2.6) at facet quadrature points and reports it with each run. The interpolation estimates of Section 5 are stated for smooth metrics, and the fixed-space perturbation bounds of Theorems 5.9, 5.10, and 5.14 hold for all three classes.

The tensor G_K may be nonconstant, anisotropic, and of variable curvature. In a periodic model two paired facets may lie on opposite sides of the displayed fundamental region; τ_F specifies their matching as for adjacent cells.

Let $\hat{V}^k$ be a reference finite element space of degree- k differential forms and let $\hat{R}_f^k$ denote its trace on facet f . For two paired facets define the reference pullback action

$$T_F^k : \text{tr } \hat{V}^k|_{\hat{F}^-} \longrightarrow \text{tr } \hat{V}^k|_{\hat{F}^+}, \quad T_F^k \eta = \tau_F^* \eta. \quad (2.3)$$

After choosing trace bases, T_F^k is a small matrix. A broken coefficient vector $c = (c_K)_K$ represents a conforming form if and only if

$$R_{F,+}^k c_{K^+} = T_F^k R_{F,-}^k c_{K^-} \quad \text{for every paired facet } F. \quad (2.4)$$

For scalar Bernstein polynomials and a simplex vertex permutation, the matrix T_F^0 is a permutation. For one- and two-forms, pullback acts on both polynomial coefficients and differential components. In general the resulting higher-order transformation matrices contain blocks beyond signed permutations [12, 36, 37].

Cell entities and their incidence are identified separately, since distinct quotient edges may have the same pair of quotient vertices (Section 4.3).

The trace-space condition is a restriction. For $\tau(t) = t + \varepsilon t(1 - t)$ with $0 < \varepsilon < 1$ the pullback of $u(t) = t$ is quadratic, so $\tau^* \mathcal{P}_1 \neq \mathcal{P}_1$.

A facet on ∂M has no partner. For scalar problems a homogeneous Dirichlet condition constrains its trace coefficients and a Neumann condition leaves them free; the relative and absolute form complexes are specified in Section 5.2.

2.1. Element matrices

Three operations act on a computation: a change of coordinates for a fixed function, a change of the local polynomial space, and a replacement of the metric. The first leaves the discrete solution unchanged, and the other two change it in ways that the analysis quantifies.

Let $F_K : \hat{K} \rightarrow M$ be an intrinsic element parametrization and let $\{\partial_i\}$ denote its coordinate tangent vectors. Write

$$G_K = (g(F_{K*} \partial_i, F_{K*} \partial_j))_{i,j=1}^n, \quad C_K = \sqrt{\det G_K} G_K^{-1}, \quad \rho_K = \sqrt{\det G_K}. \quad (2.5)$$

A scalar element routine requires only ρ_K and C_K ; a k -form routine requires ρ_K and the induced matrix on k -covectors in (4.12).

The metric compatibility condition between paired facets can be checked without an embedding. If $\tau_F : \hat{F}^+ \rightarrow \hat{F}^-$ is the facet transition, the restrictions of the two metrics to tangent vectors along the facet satisfy

$$\iota_+^* g_{K^+} = \tau_F^* \iota_-^* g_{K^-}, \quad g_{K^\pm} = F_{K^\pm}^* g, \quad (2.6)$$

Here $\iota_\pm : \hat{F}^\pm \hookrightarrow \hat{K}^\pm$ are the facet inclusions. In reference facet coordinates the right-hand side has matrix $(D\tau_F)^T (G_{K^-}|_{T\hat{F}^-} \circ \tau_F) D\tau_F$, while the left-hand side is the tangential restriction of G_{K^+} on $\hat{F}^+$. This tensor relation identifies the metric; the polynomial trace relation (2.4) identifies the finite element trace.

Consider a second parametrization $\tilde{F}_K = F_K \circ \psi$ of the same cell, with ψ a smooth diffeomorphism of reference cells. The transformed metric is

$$\tilde{G}_K(y) = D\psi(y)^T G_K(\psi(y)) D\psi(y), \quad \tilde{\rho}_K(y) = |\det D\psi(y)| \rho_K(\psi(y)). \quad (2.7)$$

If $\tilde{u} = u \circ \psi$ and $\tilde{v} = v \circ \psi$, then direct substitution yields

$$\int_{\hat{K}} \tilde{u} \tilde{v} \tilde{\rho}_K dy = \int_{\hat{K}} uv \rho_K dx, \quad (2.8)$$

$$\int_{\hat{K}} \nabla_y \tilde{u}^T \tilde{G}_K^{-1} \nabla_y \tilde{v} \tilde{\rho}_K dy = \int_{\hat{K}} \nabla_x u^T G_K^{-1} \nabla_x v \rho_K dx. \quad (2.9)$$

The same statement for differential forms follows from naturality of pullback and the coordinate invariance of the L_g^2 inner product. These identities concern the *same functions* expressed in two coordinate systems.

Proposition 2.2 (When two parametrizations give identical discrete matrices). *Suppose $\psi^* \hat{V}_p^k = \hat{V}_p^k$ and the global facet constraints are transported by the same pullback. Suppose all element integrals are exact. Then the two discrete generalized eigenproblems are related by invertible changes of coefficient basis and have identical eigenvalues, counting multiplicity.*

Proof. Let T_k be the blockwise invertible coefficient transformation representing ψ^* and restricting to a bijection between the two global constrained spaces. Equation (2.9) gives the corresponding congruences $\tilde{M}_k = T_k^T M_k T_k$ and $\tilde{A}_k = T_k^T A_k T_k$. Hence the generalized Rayleigh quotients agree after the bijection and so do their min-max values. $\square$

For a nonlinear ψ the pullback of a degree- p polynomial generally has higher degree, so two fixed-degree computations in different nonlinear charts use different conforming subspaces and agree only in the limit. The standard and twisted charts of Section 8.3 are an instance. In summary: (i) the continuous weak forms are invariant under changes of coordinates; (ii) the discrete matrices of two presentations are congruent when the change of coordinates preserves the local polynomial spaces and the facet constraints, as in Proposition 2.2; (iii) presentations with different polynomial charts give different discrete solutions that converge to the same limit. The data of the method are the element metric and the facet maps, and in this sense the method is intrinsic.

Replacing G_K changes the bilinear forms on a fixed space, changing F_K or the local polynomial family changes the space for a fixed metric, and quadrature is a third source of error. A numerical

study should distinguish

$$\underbrace{\|u - u_{h,p}(g)\|}_{\text{finite element error}}, \quad \underbrace{\|u_{h,p}(g) - u_{h,p}(\tilde{g})\|}_{\text{metric-data error}}, \quad \underbrace{\|u_{h,p}(\tilde{g}) - u_{h,p}^Q(\tilde{g})\|}_{\text{integration error}}. \quad (2.10)$$

When the metrics induce different norms, the analysis compares them by explicit tensor inequalities.

An unpaired codimension-one facet belongs to ∂M and carries the boundary trace condition. The mesh includes an orientation of each cell and subentity, which fixes the signs of integral degrees of freedom and incidence matrices. The mass and energy matrices use only the Riemannian density, so the scalar problem and the L_g^2 adjoint δ_g of d are defined on nonorientable manifolds as well; star-based formulas are read in oriented charts [3, 5]. The form experiments in this paper use orientable manifolds.

3. Bernstein–Bézier finite elements

Let Δ be the simplicial mesh of Section 2, with reference simplex $\hat{K}$ and barycentric coordinates $b_0, \dots, b_n$. The Bernstein basis of degree p is

$$B_\alpha^p(b) = \frac{p!}{\alpha_0! \cdots \alpha_n!} \prod_{i=0}^n b_i^{\alpha_i}, \quad |\alpha| = p. \quad (3.1)$$

Define the broken space

$$S_p^{-1}(\Delta) = \{s : M \rightarrow \mathbb{R} : s \circ F_K \in \mathcal{P}_p(\hat{K}) \text{ for each } K \in \Delta\}. \quad (3.2)$$

Every cell retains its own $\binom{n+p}{n}$ coefficients,

$$(s \circ F_K)(\hat{x}) = \sum_{|\alpha|=p} c_{K,\alpha} B_\alpha^p(b(\hat{x})), \quad c = (c_K)_{K \in \Delta}. \quad (3.3)$$

The coefficient $c_{K,\alpha}$ is indexed by the Bernstein domain point $\xi_{K,\alpha} = \sum_i (\alpha_i/p) v_{K,i}$ of the reference simplex.

For each paired facet the scalar trace equality is a linear functional of the two local coefficient vectors. Stack these functionals as the matrix J_0 . The conforming scalar space and a coefficient extension matrix are

$$S_p^0(\Delta) = \{s \in S_p^{-1}(\Delta) : J_0 c = 0\}, \quad c = Z_0 \hat{c}, \quad \text{im } Z_0 = \ker J_0. \quad (3.4)$$

This coefficient constraint description extends the smoothness-matrix representation of spline spaces [7, 25, 29]. For affine matching facets, continuity identifies the degree- p Bernstein trace coefficients after a vertex permutation, so Z_0 is assembled sparsely from coefficient classes. The extension matrix Z_0 depends only on the mesh and the trace conditions and is reused when the metric changes.

Figure 2 shows the trace constraint for two paired elements.

In local coordinates the scalar Laplace–Beltrami operator is

$$\Delta_g u = \frac{1}{\sqrt{\det g}} \partial_i \left(\sqrt{\det g} g^{ij} \partial_j u \right), \quad (g^{ij}) = (g_{ij})^{-1}. \quad (3.5)$$

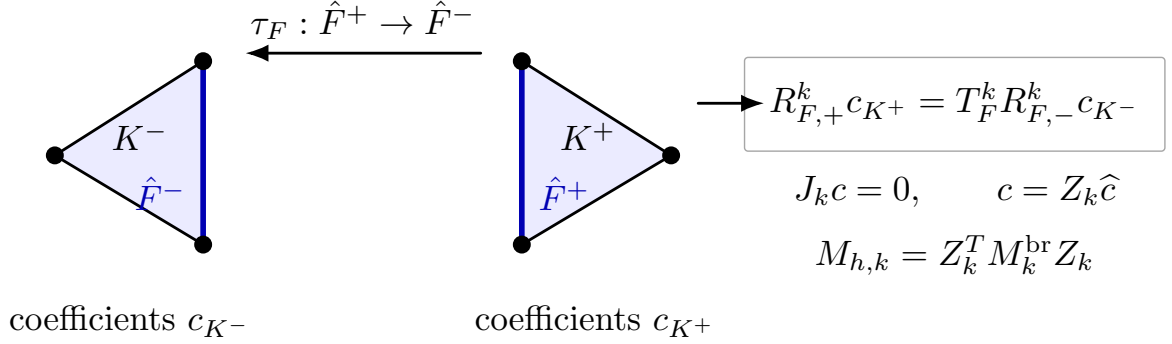


Figure 2: Trace identification of two-dimensional elements. A reference-edge pullback gives a linear constraint on the broken coefficients; its sparse extension matrix Z_k is used in global assembly. For scalar Bernstein elements, $k = 0$ and T_F^0 permutes trace coefficients.

On a connected closed manifold, the normalized Poisson problem is

$$-\Delta_g u = f, \quad \int_M u \, dV_g = 0, \quad \int_M f \, dV_g = 0. \quad (3.6)$$

The associated scalar eigenvalue problem is $-\Delta_g u = \mu u$.

For a scalar function u , denote by $\nabla_g u$ its Riemannian gradient, defined by $g(\nabla_g u, X) = X(u)$ for every tangent vector X . In local coordinates $(\nabla_g u)^i = g^{ij} \partial_j u$. The scalar energy and mass forms are

$$a_g(u, v) = \int_M g(\nabla_g u, \nabla_g v) \, dV_g, \quad b_g(u, v) = \int_M uv \, dV_g. \quad (3.7)$$

For the Poisson problem (3.6), find $u_h \in S_p^0(\Delta)$ with $b_g(u_h, 1) = 0$ such that

$$a_g(u_h, v_h) = b_g(f, v_h), \quad v_h \in S_p^0(\Delta). \quad (3.8)$$

The associated positive-shift problem replaces a_g by $a_g + \sigma b_g$ for $\sigma > 0$. On each cell, with $\rho_K = \sqrt{\det G_K}$, the element matrices are

$$(A_K)_{\alpha\beta} = \int_{\hat{K}} (\nabla B_\alpha^p)^T G_K^{-1} \nabla B_\beta^p \rho_K \, d\hat{x}, \quad (3.9)$$

$$(M_K)_{\alpha\beta} = \int_{\hat{K}} B_\alpha^p B_\beta^p \rho_K \, d\hat{x}. \quad (3.10)$$

Here ∇B_α^p is the gradient in reference coordinates. The broken matrices are $K^{\text{br}} = \text{diag}_K A_K$ and $M^{\text{br}} = \text{diag}_K M_K$. In independent coefficients the reduced matrices are

$$K_h = Z_0^T K^{\text{br}} Z_0, \quad M_h = Z_0^T M^{\text{br}} Z_0. \quad (3.11)$$

Write $\hat{c}_1$ for the global coefficient vector of the constant function, and $\hat{F} = Z_0^T F^{\text{br}}$. The mean-constrained Poisson system is equivalently expressed as

$$\begin{pmatrix} K_h & M_h \hat{c}_1 \\ \hat{c}_1^T M_h & 0 \end{pmatrix} \begin{pmatrix} \hat{c} \\ \nu \end{pmatrix} = \begin{pmatrix} \hat{F} \\ 0 \end{pmatrix}. \quad (3.12)$$

The reaction problem is $(K_h + \sigma M_h)\hat{c} = \hat{F}$. The Laplace–Beltrami spectrum is obtained from

$$K_h \hat{c} = \mu_h M_h \hat{c}. \quad (3.13)$$

The constant mode gives $\mu_{h,0} = 0$ on a connected closed manifold and for the scalar Neumann problem on a connected manifold with boundary.

Proposition 3.1 (Exact-metric Galerkin principle). *If the element tensors G_K are the pullbacks of g and the integrals (3.9)–(3.10) are evaluated exactly, then (3.11) is precisely the Galerkin restriction of the intrinsic forms (3.7) to $S_p^0(\Delta)$. The prescribed metric enters through G_K alone.*

Proof. On each element, the coordinate expression of dV_g is $\rho_K d\hat{x}$ and the gradient inner product is determined by G_K^{-1} . Summing the element integrals gives the broken forms. The columns of Z_0 extend the independent coefficients to exactly the trace-conforming coefficient vectors, so the Galerkin restrictions are $Z_0^T K^{\text{br}} Z_0$ and $Z_0^T M^{\text{br}} Z_0$. $\square$

Remark 3.2 (Metric data and quadrature). The integrands in (3.9)–(3.10) need not be polynomials even if the trial functions are. Prescribing G_K exactly at quadrature points removes geometric replacement from the formulation, but finite quadrature still contributes a consistency error. The estimate in Section 5 distinguishes these two effects.

3.1. Conformal metrics

Surface finite elements and their higher-order geometric approximation are analyzed in [15, 20, 22]; the present weak forms are evaluated using the prescribed intrinsic metric.

The earlier method [27] starts with the unit sphere and a conformal metric $\bar{g} = e^{2\lambda} g_S$. The metric is represented by the positive scalar weight $w = e^{2\lambda}$, and the spline space is built from homogeneous spherical Bernstein–Bézier polynomials. It is useful to record exactly which part of that construction extends to our general intrinsic setting.

Proposition 3.3 (The conformal reduction in dimension two). *Let (M, g_0) be an oriented Riemannian manifold of dimension n and let $g = e^{2\lambda} g_0$ for a smooth function λ . For scalar functions u, v ,*

$$b_g(u, v) = \int_M e^{n\lambda} uv \, dV_{g_0}, \quad (3.14)$$

$$a_g(u, v) = \int_M e^{(n-2)\lambda} g_0(\nabla_{g_0} u, \nabla_{g_0} v) \, dV_{g_0}. \quad (3.15)$$

In particular, when $n = 2$ and $w = e^{2\lambda}$,

$$a_g = a_{g_0}, \quad b_g(u, v) = \int_M w uv \, dV_{g_0}. \quad (3.16)$$

The scalar finite element matrices computed with the exact metrics g and g_0 therefore have the same stiffness matrix for any fixed conforming trial space in dimension two; their mass matrices differ by the weight w .

Proof. In local coordinates, $g^{-1} = e^{-2\lambda} g_0^{-1}$ and $\sqrt{\det g} = e^{n\lambda} \sqrt{\det g_0}$. Substitution into the two scalar bilinear forms proves all the statements. $\square$

For the unit two-sphere, the round metric in spherical coordinates is $g_{S^2} = d\theta^2 + \sin^2 \theta d\varphi^2$; the conformal metric in [27] is $g = e^{2\lambda} g_{S^2}$. The proposition gives the weighted weak equations of [27]. That paper uses homogeneous spherical Bernstein polynomials, while the present formulation uses ordinary reference-simplex Bernstein polynomials with their associated trace constraints. Our $S_p^0(\Delta)$ contains the constant function for every p because $\sum_{|\alpha|=p} B_\alpha^p = 1$ on each reference simplex. The even-degree requirement for the constants and exact spherical-harmonic reproduction in [27] apply to its homogeneous spherical polynomial space.

In dimension three, a conformal metric changes the stiffness matrix through the factor e^λ and the mass matrix through $e^{3\lambda}$. More generally, a nonconformal metric is carried by the full matrix $G_K^{-1} \rho_K$ in (3.9).

4. Compatible finite element spaces

The edge-level differentiation and smoothness identities for constrained Bernstein vector splines on triangulated surfaces are developed in [25]. The present construction uses the same coefficient-level conformity principle with reference-facet pullbacks for k -forms in arbitrary dimension.

Differential forms provide an intrinsic finite element language for gradient, curl, divergence and Hodge-type systems in every dimension. They provide compatible finite element spaces assembled by the same constraint and sparse-matrix construction. Let $\Lambda^k(M)$ denote differential k -forms and $d : \Lambda^k(M) \rightarrow \Lambda^{k+1}(M)$ the exterior derivative. Its coefficient matrix on each cell does not depend on g , whereas L^2 inner products and adjoints do.

For $r \geq 1$, put $\hat{V}_r^k = \mathcal{P}_r^- \Lambda^k(\hat{K})$, $0 \leq k \leq n$. One algebraic construction is

$$\mathcal{P}_r^- \Lambda^k(\hat{K}) = \mathcal{P}_{r-1} \Lambda^k(\hat{K}) + \kappa \mathcal{H}_{r-1} \Lambda^{k+1}(\hat{K}), \quad (4.1)$$

where κ denotes Koszul contraction by the coordinate vector and $\mathcal{H}_j$ denotes homogeneous polynomial forms of degree j [3, 4, 6, 28]. The sum in (4.1) is direct: the first summand has polynomial degree at most $r-1$, whereas the second consists of homogeneous forms of degree r . The spanning generators for the image of κ are linearly dependent, since κ is not injective on $\mathcal{H}_{r-1} \Lambda^{k+1}$. The resulting sequence of spaces is a complex. In three dimensions its local dimensions for $r = 1, 2, 3$ are respectively

$$(4, 6, 4, 1), \quad (10, 20, 15, 4), \quad (20, 45, 36, 10). \quad (4.2)$$

The form spaces carry their own local bases. Our reference implementation constructs the span in monomial/Koszul coordinates and then applies a Bernstein-coordinate basis change for improved local conditioning. The final degrees of freedom are associated with vertices, edges, faces, or cell interiors as appropriate for k and r .

4.1. Local bases and facet transformations

For the triangle with barycentric coordinates b_0, b_1, b_2 , the lowest-order sequence has dimensions $(3, 3, 1)$. Its scalar basis is (b_0, b_1, b_2) ; a basis for $\mathcal{P}_1^- \Lambda^1$ is given by the Whitney edge forms

$$w_{ij} = b_i db_j - b_j db_i, \quad (i, j) = (0, 1), (0, 2), (1, 2). \quad (4.3)$$

The top-degree space is generated by $db_1 \wedge db_2$. For the tetrahedron, the corresponding local dimensions are $(4, 6, 4, 1)$. Its lowest-order two-forms can be generated by the four Whitney face forms

$$w_{ijk} = b_i db_j \wedge db_k - b_j db_i \wedge db_k + b_k db_i \wedge db_j, \quad 0 \leq i < j < k \leq 3. \quad (4.4)$$

On a tetrahedral face the tangential traces of the edge forms span $\mathcal{P}_1^-\Lambda^1$ on the triangle; the tangential traces of the face forms span its one-dimensional $\mathcal{P}_1^-\Lambda^2$ space. The interior components that vanish under trace do not enter a facet constraint. The higher-order trace dimensions on a triangular face are $(6, 8, 3)$ for $r = 2$ and $(10, 15, 6)$ for $r = 3$ in form degrees $0, 1, 2$, respectively [3, 6].

The Bernstein connection can be made explicit for all $r \geq 1$. If $B_\alpha^{r-1}(b)$ is the barycentric Bernstein basis of degree $r - 1$, then the collection

$$\{B_\alpha^{r-1}(b)w_{i_0\dots i_k} : |\alpha| = r - 1, 0 \leq i_0 < \dots < i_k \leq n\} \quad (4.5)$$

spans $\mathcal{P}_r^-\Lambda^k$; a linearly independent basis is obtained by eliminating its algebraic dependencies. The Whitney zero-forms are the barycentric coordinates. In three dimensions at $r = 2, k = 1$, the twenty-dimensional space is spanned by the twenty-four products $b_\ell w_{ij}$, $0 \leq \ell \leq 3, i < j$. The local entity moments select independent global degrees of freedom from this span; the Bernstein change of basis affects local matrices and conditioning, while the trace constraint is imposed on the complete form-valued polynomial trace [2, 6, 28].

For example, consider the reference-facet permutation $\tau(b_0, b_1, b_2) = (b_0, b_2, b_1)$. On the ordered scalar basis and ordered edge-form basis in (4.3), its pullback matrices are

$$T_F^0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad T_F^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad T_F^2 = (-1). \quad (4.6)$$

The minus sign records the reversal of the oriented edge $(1, 2)$ and of the oriented face. For a general affine face map $x = Ay + b$, each polynomial coefficient is composed with $Ay + b$ and each coordinate k -differential transforms by the $k \times k$ minors of A . This gives the higher-order matrices T_F^k in the chosen trace basis. The next subsection uses them in the constrained assembly.

For an affine reference change of variables τ , polynomial-form pullback acts both on coefficients and on differentials. It satisfies

$$d\tau^* = \tau^*d. \quad (4.7)$$

Accordingly, the facet transformations are the reference pullbacks of polynomial differential forms.

Define the broken space $W_{h,r}^k = \prod_{K \in \Delta} F_K^{-1*} \hat{V}_r^k$ and let D_k^{br} be the block diagonal local exterior-derivative matrix. Let J_k be the constraint matrix whose rows are the paired-facet constraints (2.4). We write

$$V_{h,r}^k = \ker J_k, \quad Z_k : \mathbb{R}^{N_k} \longrightarrow \mathbb{R}^{N_k^{\text{br}}}, \quad \text{im } Z_k = \ker J_k, \quad (4.8)$$

where Z_k has full column rank. We use the same letter Z as for the scalar Bernstein constraint reduction; here its columns are assembled from compatible entity moments. Its columns are the broken coefficient vectors obtained from the independent global degrees of freedom. A sparse entity-based implementation constructs Z_k directly; a dense nullspace computation gives an independent small-mesh reference. Throughout, Z_k is called the *coefficient-extension matrix*; its image is the constrained coefficient space.

Theorem 4.1 (Constrained assembly preserves the differential complex). *Suppose the local spaces are closed under d , their facet traces agree through the pullback maps (2.3), and the constraint matrices are exact. Then*

$$D_k^{\text{br}} \text{im } Z_k \subseteq \text{im } Z_{k+1}. \quad (4.9)$$

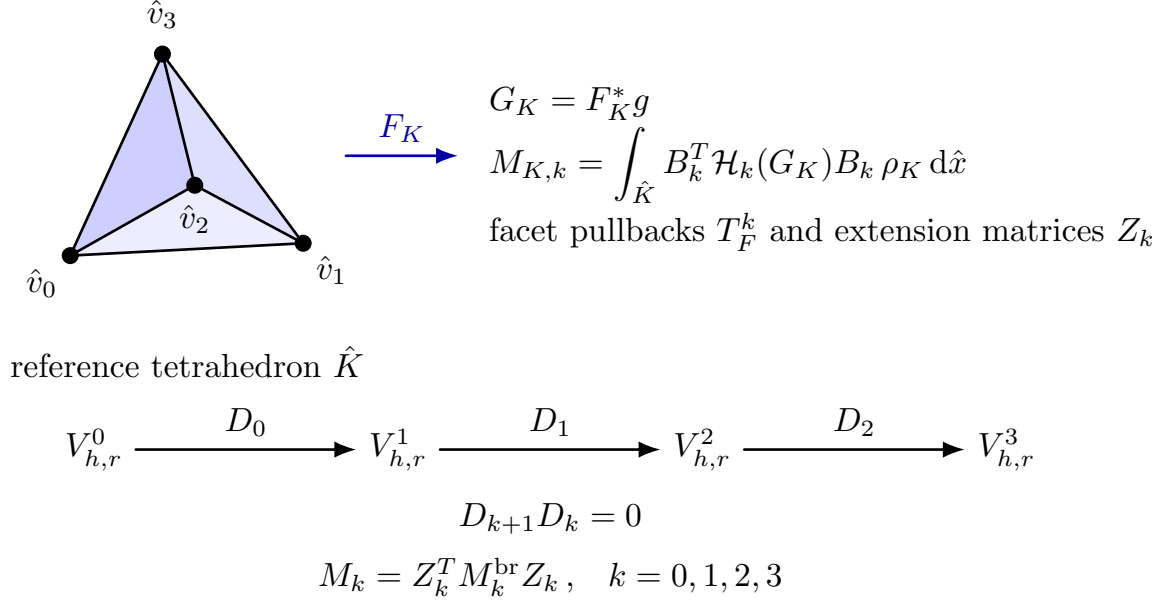


Figure 3: Three-dimensional reference element and global compatible finite element sequence. The differential matrices D_k are determined by the reference spaces and facet constraints in a fixed basis; the metric enters through the local Hodge matrices $M_{K,k}$.

For any left inverses $Q_{k+1} Z_{k+1} = I$, define

$$D_k = Q_{k+1} D_k^{\text{br}} Z_k. \quad (4.10)$$

The matrices D_k represent the exterior derivative on the conforming spaces; in particular,

$$D_k^{\text{br}} Z_k = Z_{k+1} D_k, \quad D_{k+1} D_k = 0. \quad (4.11)$$

Proof. Let $u \in \text{im } Z_k$. On a paired facet, its two tangential traces agree after pullback. Taking the exterior derivative along the facet and applying (4.7) shows that the two tangential traces of du also agree. Hence $du \in \text{im } Z_{k+1}$, proving (4.9). The left-inverse identity and invariance give $Z_{k+1} Q_{k+1} D_k^{\text{br}} Z_k = D_k^{\text{br}} Z_k$. Applying the next local differential and using $(D_{k+1}^{\text{br}} D_k^{\text{br}}) = 0$ gives $Z_{k+2} D_{k+1} D_k = 0$. Since Z_{k+2} is injective, $D_{k+1} D_k = 0$. $\square$

The tetrahedral instance of the assembled sequence is illustrated in Figure 3.

Remark 4.2 (Global derivative matrix). Fix an independent global coefficient basis in each degree and let $Q_{k+1} = (Z_{k+1}^T Z_{k+1})^{-1} Z_{k+1}^T$ in the Euclidean broken coefficient inner product. Then $D_k = Q_{k+1} D_k^{\text{br}} Z_k$ is its unique coefficient representation. Any other left inverse produces the same matrix on $D_k^{\text{br}} \text{im } Z_k$, because that range lies in $\text{im } Z_{k+1}$. A change of global coefficient basis changes the matrix of D_k and preserves the intertwining identity $D_k^{\text{br}} Z_k = Z_{k+1} D_k$. Galerkin bilinear forms use Z_k^T on test coefficients, and the extraction of D_k uses the left inverse Q_{k+1} .

Let $\{\omega_{K,i}^k\}$ be the chosen reference basis expressed as covariant k -forms in the local coordinates. For strictly increasing multi-indices $I = (i_1 < \dots < i_k)$ and $J = (j_1 < \dots < j_k)$, write $(G_K^{-1})_{I,J}$ for the submatrix with these ordered rows and columns, and set $(G_K^{-1})_{\emptyset,\emptyset} = 1$. All coefficient vectors

use this increasing-index convention; signs under a facet permutation occur in its pullback matrix. The induced coefficient matrix for the inner product of k -covectors is

$$\mathcal{H}_k(G_K)_{I,J} = \det((G_K^{-1})_{I,J}). \quad (4.12)$$

For example, when $G_K = \text{diag}(a, b, c)$ in dimension three, the ordered two-form basis $(dx^1 \wedge dx^2, dx^1 \wedge dx^3, dx^2 \wedge dx^3)$ has $\mathcal{H}_2(G_K) = \text{diag}((ab)^{-1}, (ac)^{-1}, (bc)^{-1})$. This simple diagonal case checks the form-component ordering in the local implementation. If $B_k(\hat{x})$ collects the coefficient vectors of the reference basis, the local k -form mass matrix is

$$M_{K,k} = \int_{\hat{K}} B_k(\hat{x})^T \mathcal{H}_k(G_K(\hat{x})) B_k(\hat{x}) \rho_K(\hat{x}) d\hat{x}. \quad (4.13)$$

Conforming assembly gives

$$M_k = Z_k^T \text{diag}_K(M_{K,k}) Z_k. \quad (4.14)$$

If the local reference basis is converted to entity moments by $L_{K,k}$, the block in (4.14) is $L_{K,k}^{-T} M_{K,k} L_{K,k}^{-1}$, and Z_k acts on entity-moment coefficients. The transformation must be applied consistently to the differential and to all form matrices.

The discrete codifferential is the adjoint $\delta_{h,k} = M_{k-1}^{-1} D_{k-1}^T M_k$. Consequently the symmetric matrix of the discrete Hodge quadratic form on k -forms is

$$\begin{aligned} A_k &= D_k^T M_{k+1} D_k \\ &\quad + M_k D_{k-1} M_{k-1}^{-1} D_{k-1}^T M_k, \end{aligned} \quad (4.15)$$

with the nonexisting term suppressed for $k = 0$ or $k = n$. The generalized eigenproblem is $A_k u = \mu M_k u$. The metric affects M_k , and therefore the codifferential and spectrum; the differential matrices D_k are unchanged when the same finite element mesh and facet constraints are used with a different compatible metric.

Proposition 4.3 (Harmonic dimension at fixed resolution). *For positive-definite matrices M_k and a finite-dimensional complex (4.11),*

$$\dim \ker A_k = \dim \ker D_k - \text{rank } D_{k-1}, \quad (4.16)$$

with the evident endpoint conventions. The number in (4.16) is independent of the metric matrices.

Proof. Taking the quadratic form of (4.15) shows that $\ker A_k = \ker D_k \cap (\text{im } D_{k-1})^{\perp_{M_k}}$. By (4.11), $\text{im } D_{k-1} \subseteq \ker D_k$. Orthogonal decomposition of $\ker D_k$ by this subspace gives (4.16). $\square$

The proposition identifies discrete harmonic dimension with the cohomology of the assembled finite-dimensional complex. Proposition 5.6 relates this dimension to the Betti numbers of M under its approximation hypotheses.

4.2. Global assembly

The entity-based decomposition and canonical moment degrees of freedom follow the standard polynomial differential-form construction [3, 4, 6]. We use two bases of the same local space: an algebraic basis of polynomial forms, which is convenient for d , pullbacks, and metric matrices, and the basis dual to degrees of freedom on vertices, edges, faces, and cell interiors, which is convenient for sparse assembly.

The trimmed polynomial spaces $\hat{V}_r^k$ and the Koszul representation are defined in (4.1). The Koszul generators and the spanning set (4.5) are linearly dependent. A numerical basis is selected by a rank-revealing factorization of the spanning columns or from an explicit Bernstein-associated basis. The dimension is

$$\dim \mathcal{P}_r^- \Lambda^k(\hat{K}) = \binom{r+k-1}{k} \binom{n+r}{n-k}. \quad (4.17)$$

In coordinates $x = (x_1, \dots, x_n)$, store a monomial differential form as a list of terms $c_{\alpha, I} x^\alpha dx_{i_1} \wedge \dots \wedge dx_{i_k}$. Then

$$d(x^\alpha dx_I) = \sum_{j \notin I} \alpha_j x^{\alpha - e_j} dx_j \wedge dx_I, \quad (4.18)$$

with the usual wedge sign, which gives rational local derivative matrices before any floating-point change of basis.

For a subsimplex $f \subseteq \hat{K}$ of dimension $d \geq k$, the canonical degrees of freedom can be taken in the form

$$\ell_{f, \eta}(\omega) = \int_f \text{tr}_f \omega \wedge \eta, \quad \eta \in \mathcal{P}_{r+k-d-1} \Lambda^{d-k}(f), \quad (4.19)$$

where a negative polynomial index supplies no degrees of freedom. The integrand has degree d , so the orientation of f fixes the sign of the moment. The collection of degrees of freedom over all entities is unsolvent for the trimmed space, with the standard finite-element moment conventions [3, 4, 6]. For $r > 1$ the edge and face moments carry several polynomial components, so the forms cannot be assembled as unrelated scalar fields.

Let $\{\varphi_i^{K,k}\}_{i=1}^{N_{K,k}}$ be any local algebraic basis and let $L_{K,k}$ be its degree-of-freedom matrix,

$$(L_{K,k})_{ai} = \ell_a(\varphi_i^{K,k}). \quad (4.20)$$

If $L_{K,k}$ is invertible, the coefficients in the nodal/moment dual basis are related to the algebraic coefficients by $z_K = L_{K,k} c_K$. Consequently the local mass and differential matrices in moment coordinates are

$$M_{K,k}^{\text{mom}} = L_{K,k}^{-T} M_{K,k}^{\text{alg}} L_{K,k}^{-1}, \quad D_{K,k}^{\text{mom}} = L_{K,k+1} D_{K,k}^{\text{alg}} L_{K,k}^{-1}. \quad (4.21)$$

The derivative is transformed on both sides; omitting either factor destroys the complex identities.

Let a paired facet map be affine in reference coordinates, $x = Ay + b$. The pullback of a monomial form is obtained by composing x^α with $Ay + b$ and evaluating the $k \times k$ minors of A in the differential components:

$$(Ay + b)^\alpha (A^* dx_I) = \sum_{|J|=k} (Ay + b)^\alpha \det(A_{I,J}) dy_J. \quad (4.22)$$

The resulting reference matrix maps the trace coefficients on one side to those on the other, consistently with the orientation convention in (4.19).

For vertex-permutation maps A , the pullback preserves the chosen polynomial spaces and satisfies the group law. In the scalar Bernstein basis this action permutes the multi-indices, but the corresponding moment transformations for high-order forms are generally small matrices. At the reference level we require the commuting identity

$$D_{F,k} T_F^k = T_F^{k+1} D_{F,k}, \quad (4.23)$$

with the reference-facet differentials. It follows from naturality, and its numerical residual is a check of orientation and indexing.

Start with the disjoint union of all cells and carry identifiers for their vertices, oriented edges, oriented faces, and interiors. Each facet pairing induces subentity correspondences. Form their equivalence classes while retaining the attaching maps and the orientation/polynomial transformation on each class. When an entity is related to itself by a nontrivial transformation, its admissible degrees of freedom are the invariant subspace of that transformation.

A cell may also meet itself across a quotient facet. Its trace condition is an ordinary row of J_k , an orientation-reversing pairing may annihilate some trace components, and Z_k records the remaining independent degrees of freedom.

With moment-space broken matrices, the global matrices have the familiar finite element form

$$M_{h,k} = Z_k^T \text{diag}_K(M_{K,k}^{\text{mom}}) Z_k. \quad (4.24)$$

The matrix Z_k may contain small transformation blocks associated with form traces. The global derivative $D_{h,k}$ is determined uniquely by the intertwining equation

$$D_k^{\text{br}} Z_k = Z_{k+1} D_{h,k}. \quad (4.25)$$

The left inverse $Q_{k+1} = (Z_{k+1}^T Z_{k+1})^{-1} Z_{k+1}^T$ extracts $D_{h,k}$ from this identity, and the Hodge–Laplace matrix is formed as in (4.14).

Proposition 4.4 (Independence of the global coefficient basis). *Let $\tilde{Z}_k = Z_k S_k$ with S_k invertible. If $\tilde{D}_k = S_{k+1}^{-1} D_k S_k$ and $\tilde{M}_k = S_k^T M_k S_k$, then the finite element functions, discrete cohomology dimensions, and generalized Hodge spectra are unchanged.*

Proof. The ranges of Z_k and $\tilde{Z}_k$ coincide, hence they represent the same conforming functions. Direct substitution gives $\tilde{D}_{k+1} \tilde{D}_k = S_{k+2}^{-1} D_{k+1} D_k S_k = 0$ and a chain isomorphism via the maps S_k . The Hodge quadratic and mass forms are pulled back by the same invertible coefficient changes, so their generalized Rayleigh quotients and min–max values agree. $\square$

Entity assembly scales with the number of mesh entities, and the local reference matrices and facet transformations are reused for all elements with the same degree and facet permutation. The conditioning of the reference moment matrices is reported in Table 9.

4.3. Quotient cell incidence

The smallest periodic example illustrates why a global mesh is more than its quotient vertex set. Represent a circle by an interval with its two endpoints identified. The quotient has one vertex v and one oriented edge e , with

$$\partial_1 e = v - v = 0. \quad (4.26)$$

A reconstruction that discards edges joining a vertex to itself would replace the circle by a point. A periodic square cell structure of the torus has one vertex, two distinct oriented loop edges a, b , and one face whose attaching path is $aba^{-1}b^{-1}$; its cellular boundary vanishes over the real numbers. It has the expected cohomology dimensions $(1, 2, 1)$.

The same occurs for three-dimensional quotient meshes: a cell may pair two of its own facets, and distinct edges may share their endpoint classes. The entity construction therefore records local oriented incidences first and then applies the face pairings to those entities.

For the five-cell spherical quotient used in Section 8.2, the observed global form dimensions at $r = 1$ are $(1, 6, 10, 5)$ and the computed harmonic dimensions are $(1, 0, 0, 1)$. Rank-nullity and the cochain identities then give

$$(\text{rank } D_0, \text{rank } D_1, \text{rank } D_2) = (0, 6, 4). \quad (4.27)$$

At $r = 2$ the same calculation from global dimensions $(7, 32, 45, 20)$ gives $(6, 26, 19)$, and at $r = 3$ the dimensions $(23, 93, 120, 50)$ give $(22, 71, 49)$. The number of degrees of freedom changes with the order while the assembled cohomology stays fixed.

Correct dimensions and incidence ranks check the quotient entity map; stability is the separate matter of Section 4.4.

4.4. Commuting projections

Convergence of the constrained polynomial complex is established using commuting projections with uniform graph-norm bounds. We specify the projection assumptions in terms of the element maps and the trace operators used in global assembly.

Let $H\Lambda^k(M)$ denote the graph space of square-integrable differential k -forms whose exterior derivative is square integrable. On a compact manifold this is a Hilbert space with norm

$$\|u\|_{H\Lambda^k(M,g)}^2 = \|u\|_{L_g^2}^2 + \|du\|_{L_g^2}^2. \quad (4.28)$$

A suitable commuting projection has the properties

$$\pi_h^k : H\Lambda^k(M) \rightarrow V_{h,r}^k, \quad \pi_h^k|_{V_{h,r}^k} = I, \quad d\pi_h^k = \pi_h^{k+1}d, \quad (4.29)$$

with an operator norm bounded independently of the mesh size, and of the degree for an hp statement.

In broken coefficients, a candidate elementwise interpolant π_K^k must satisfy

$$R_{F,+}^k \pi_{K,+}^k u = T_F^k R_{F,-}^k \pi_{K,-}^k u \quad \text{for each } u \in H\Lambda^k(M). \quad (4.30)$$

Commuting projections built on mesh-patch regularization are developed in [5, 19, 34, 35].

Proposition 4.5 (Transfer of intrinsic commuting projections). *Let M be a compact smooth Riemannian manifold equipped with a family of conforming simplicial triangulations, smooth uniformly regular element charts, and affine reference-facet transition maps. Suppose this family satisfies the uniform chart, shape-regularity, mesh-patch and regularization hypotheses of a bounded commuting-projection theorem for intrinsic polynomial FEEC spaces, such as [34, 35]. Let $V_{h,r}^k$ be assembled by the exact trace constraints (2.4) from the pulled-back spaces $\mathcal{P}_r^- \Lambda^k$. Then these constrained spaces coincide with the intrinsic FEEC spaces on that triangulation. In particular, the commuting projections furnished by the cited theorem have range $V_{h,r}^k$ and satisfy (4.29) with the bounds of that theorem.*

Proof. The local reference spaces agree by construction. On each identified facet the intrinsic tangential trace is the pullback under the facet map. Equality of these traces is exactly the matrix relation (2.4), independent of the choice of local basis or of whether Z_k is obtained by sparse entity numbering or by a nullspace basis. Consequently the global constrained space is the ordinary conforming intrinsic finite element space on the given triangulation. Apply the uniformly bounded commuting projection for that mesh family. Its range, retraction property and commutation with d are preserved by this identification. Uniform equivalence of the metric-dependent and chartwise graph norms follows from the stated chart and metric regularity. $\square$

The uniform bounds rest on the mesh-patch assumptions of the cited theorem. The identification of spaces is also visible in the dimensions. When no entity is carried to itself by a nontrivial transformation, as for a free quotient, the degrees of freedom (4.19) give

$$\dim V_{h,r}^k = \sum_{d=k}^n N_d \binom{r+k-1}{d} \binom{d}{k}, \quad (4.31)$$

where N_d is the number of d -dimensional entities of the quotient; this is the dimension of the intrinsic space $\mathcal{P}_r^- \Lambda^k$ on the same cell complex [4]. Every assembled dimension in Table 10 and Table 20 equals this count; for the 60-cell Seifert–Weber mesh $(N_0, N_1, N_2, N_3) = (8, 68, 120, 60)$, and $r = 2, k = 1$ gives $2 \cdot 68 + 2 \cdot 120 = 376$.

For finite isometric quotient meshes, a commuting projection on the covering mesh yields a commuting projection on the quotient by averaging.

A quotient mesh in which distinct cells or edges share their vertex sets is a Δ -complex. The five-cell S^3/Γ mesh has one vertex, and the small periodic-torus mesh has several edges with the same endpoints. Proposition 4.5 therefore applies to a covering triangulation, and Theorem 4.6 carries the projection to these quotients: the spherical example is a finite isometric quotient, and the periodic torus at a fixed resolution is a finite translation quotient of a finer periodic mesh.

Theorem 4.6 (Averaged commuting projection on a finite quotient). *Let $\widetilde{M}$ be a compact Riemannian manifold on which a finite group Γ acts freely by isometries, and let $M = \widetilde{M}/\Gamma$. Suppose a Γ -invariant simplicial finite element mesh and a form subcomplex $\widetilde{V}_h^\bullet \subset H\Lambda^\bullet(\widetilde{M})$ are given, with the pullbacks γ^* preserving each $\widetilde{V}_h^k$. Suppose there are uniformly bounded commuting projections $\widetilde{\pi}_h^k$ onto these spaces. Define*

$$\widetilde{\pi}_{h,\Gamma}^k = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} (\gamma^*)^{-1} \widetilde{\pi}_h^k \gamma^*. \quad (4.32)$$

Then $\widetilde{\pi}_{h,\Gamma}^k$ is a commuting projection onto $\widetilde{V}_h^k$, is equivariant under the group action, and satisfies

$$\|\widetilde{\pi}_{h,\Gamma}^k\|_{H\Lambda^k \rightarrow H\Lambda^k} \leq \|\widetilde{\pi}_h^k\|_{H\Lambda^k \rightarrow H\Lambda^k}. \quad (4.33)$$

It restricts to the invariant subcomplex and therefore induces a uniformly bounded commuting projection on the quotient finite element spaces.

Proof. Each summand maps into $\widetilde{V}_h^k$ because the mesh and subcomplex are invariant. If $v_h \in \widetilde{V}_h^k$, then $\gamma^* v_h$ lies in the same space, and $\widetilde{\pi}_h^k \gamma^* v_h = \gamma^* v_h$. Hence every summand in (4.32) fixes v_h , proving that the average is a projection. Pullback commutes with the exterior derivative and $\widetilde{\pi}_h^\bullet$ commutes with it by hypothesis, so the average also commutes. Reindexing the finite sum shows that it commutes with every γ^* . Each group pullback is an isometry for the L^2 and graph norms because the action preserves g , giving (4.33) by the triangle inequality. The pullback by the covering map identifies quotient $H\Lambda^k(M)$ with invariant covering-space forms. The covering map has degree $|\Gamma|$, so both L^2 and graph norms of a pulled-back form are multiplied by $|\Gamma|^{1/2}$; this common factor cancels in the operator-norm estimate for the descended projection. The invariant projection therefore descends with the same bound. $\square$

Remark 4.7. The averaged projection is a global analytical operator. Its norm is bounded by the covering-space projection norm, and its domain is the finite isometric quotient setting of Theorem 4.6.

Let $\mathcal{C}_h^\bullet$ denote the oriented simplicial cochain complex of a genuine triangulation of M , formed with all quotient entities and their attaching maps. The Whitney map $W_h : \mathcal{C}_h^k \rightarrow V_{h,r}^k$ and integration map R_h satisfy the cochain relations

$$dW_h = W_h \partial^*, \quad R_h d = \partial^* R_h, \quad R_h W_h = I \quad (4.34)$$

under the conventional orientation and normalization, which identifies the Whitney subcomplex with simplicial cochains.

Under (4.29) a discrete form that is exact in the continuous complex is discretely exact, and Whitney forms represent the continuous cohomology classes; Proposition 5.6 combines these two facts.

We compute the following three matrix residuals in compatible norms:

$$\begin{aligned} & \|J_k Z_k\|, \\ & \|D_k^{\text{br}} Z_k - Z_{k+1} D_k\|, \\ & \|D_{k+1} D_k\|. \end{aligned} \quad (4.35)$$

They measure trace conformity, invariance under the exterior derivative, and the complex property. The quotient-cell cohomology is an independent check of the entity identifications.

5. Approximation and spectral analysis

We distinguish the error of the finite element approximation in the prescribed Riemannian metric from the error arising when the metric or element integrals are replaced [10, 23, 24]. This permits general intrinsic metrics while retaining familiar estimates from finite element analysis.

Let h_K denote the intrinsic diameter of K , and let $h = \max_K h_K$. Assume a family of compatible intrinsic triangulations with uniformly shape-regular reference charts in scaled local coordinates. We use $\|\cdot\|_{m,g}$ to denote the Sobolev norm defined by the metric and any fixed smooth partition of unity. On a compact manifold these norms are equivalent to the usual local-chart Sobolev norms.

Theorem 5.1 (Scalar source and eigenvalue estimates). *Let (M, g) be compact, smooth and without boundary, and let the intrinsic charts be shape regular with uniformly controlled derivatives through order $p+1$. Assume compatible scalar trace spaces, $p+1 > n/2$, and exact integration. Let $I_h : H^{p+1}(M) \rightarrow S_p^0(\Delta)$ be the conforming interpolant constructed in Lemma 5.2 under its scaled-chart and facet hypotheses. The interpolant of Lemma 5.2 satisfies*

$$\|u - I_h u\|_{1,g} \leq Ch^p \|u\|_{p+1,g}. \quad (5.1)$$

For the exact-metric Galerkin solution of (3.8),

$$\|u - u_h\|_{1,g} \leq Ch^p \|u\|_{p+1,g}. \quad (5.2)$$

If the adjoint problem has H^2 regularity, the Aubin–Nitsche duality argument with (5.1) at $p=1$ gives

$$\|u - u_h\|_{0,g} \leq Ch^{p+1} \|u\|_{p+1,g}. \quad (5.3)$$

For a positive eigenvalue μ whose eigenspace E_μ lies in $H^{p+1}(M)$, the conforming scalar Galerkin eigenvalues in the associated cluster satisfy, for h sufficiently small,

$$0 \leq \mu_{h,j} - \mu \leq C_\mu h^{2p}, \quad (5.4)$$

provided the eigenspace is separated from the rest of the spectrum. Constants may depend on g , the atlas regularity, p , and the spectral gap, but not on h .

Proof. By Proposition 3.1, the discrete bilinear form is the restriction of the coercive continuous form. Céa's lemma and (5.1) give (5.2). The usual duality argument on the compact manifold, with H^2 regularity of the adjoint resolvent, yields (5.3). The Laplacian has compact resolvent on a compact manifold. Applying the conforming selfadjoint spectral approximation theorem to the approximate eigenspaces gives an eigenvalue error bounded by the square of their energy best-approximation error [8, 13]; the min-max principle gives the one-sided inequality. $\square$

For numerical quadrature, these exact-integration estimates acquire the additional consistency contribution controlled by Proposition 5.3, when its regularity and relative-form hypotheses hold.

The scalar estimates above are the standard conforming Galerkin and compact-operator consequences of the interpolation hypotheses [8, 13, 16, 17].

Lemma 5.2 (Local approximation in uniformly regular charts). *Fix $p \geq 1$ and assume $p+1 > n/2$, so the point values defining the nodal interpolant are continuous functionals on H^{p+1} . Suppose the scaled element charts F_K and their inverses have uniformly bounded derivatives through order $p+1$, the pullback metrics satisfy Assumption 2.1, and the affine reference-facet maps identify the scalar interpolation degrees of freedom. Then there is a conforming interpolant I_h into $S_p^0(\Delta)$ such that*

$$\|u - I_h u\|_{H^m(K,g)} \leq Ch_K^{p+1-m} \|u\|_{H^{p+1}(K,g)}, \quad m = 0, 1,$$

The constant depends on the chart and metric regularity and on p , but is independent of h .

Proof. Pull back the function to the reference simplex. Bramble–Hilbert and nodal polynomial interpolation there give the degree- p estimates under $p+1 > n/2$; the chain rule and the scaled chart bounds transfer these to the intrinsic Sobolev norms and supply the powers of h_K . On a common facet the two element traces are represented in the same affine reference-facet coordinates. Identifying their trace degrees of freedom therefore gives the same polynomial trace, and the local interpolants assemble into a conforming function. Summation over the elements gives the global estimate [16, 17, 29]. $\square$

Proposition 5.3 (Consistency of tensor-product Duffy quadrature). *Let $D : [0, 1]^n \rightarrow \hat{K}$ be a Duffy map, let f_K be an element integrand on $\hat{K}$, and define its cube integrand by $F_K(t) = f_K(D(t))|\det DD(t)|$. Let $\mathcal{Q}_q$ denote the q -point tensor Gauss rule applied to F_K . Assume the transformed integrand F_K extends to a C^{2q} function on a neighborhood of the closed cube $[0, 1]^n$. In particular, it suffices that f_K admit a C^{2q} extension to a neighborhood of the closed reference simplex (for metric integrands, with the extended tensor remaining positive definite). Then*

$$\left| \int_{\hat{K}} f_K \, d\hat{x} - \mathcal{Q}_q(F_K) \right| \leq C_{n,q} \max_{1 \leq j \leq n} \|\partial_{t_j}^{2q} F_K\|_{L^\infty([0,1]^n)}.$$

If the resulting assembled quadrature mass and stiffness forms differ from the exact forms by relative amounts at most $0 \leq \eta_M, \eta_A < 1$ on the fixed constrained space, respectively, then every positive scalar eigenvalue obeys

$$\frac{1 - \eta_A}{1 + \eta_M} \leq \frac{\mu_{h,j}^{\mathcal{Q}}}{\mu_{h,j}} \leq \frac{1 + \eta_A}{1 - \eta_M}.$$

Proof. Apply the one-dimensional Gaussian quadrature remainder successively in each tensor coordinate, using the positivity and unit sums of the other quadrature weights. The Duffy map

and its Jacobian are polynomial in the cube variables. The Jacobian vanishes on parts of the cube boundary and multiplies an integrand that extends smoothly to the closed simplex, so the transformed integrand is C^{2q} up to those faces. The second assertion follows by bounding each generalized Rayleigh quotient and using the min–max principle. When the relative bilinear-form bounds vanish under refinement, standard Strang consistency estimates apply to the source problem [17]. $\square$

The extension hypothesis holds for every chart used in Sections 7 and 8. The torus, tube, and tube-surface charts are affine with smooth, uniformly positive-definite coordinate metrics. The radial charts $y \mapsto y/|y|$ on S^3/Γ and $y \mapsto y/r$ on the Seifert–Weber space are real-analytic on a neighborhood of the closed reference simplex, because $|y|$ and r are bounded away from zero on the convex hull of the cell vertices, and the pullback tensors (8.10) and (8.25) are positive definite there. The Duffy Jacobian vanishes on faces of the cube and multiplies a smooth integrand, so the transformed integrand is smooth up to the boundary of the cube. The measured chart constants are in Table 11, and the computed values are in addition insensitive to the Gauss order (Sections 8.5 and 8.7).

Remark 5.4 (Quadrature order and polynomial degree). For a polynomial scalar integrand of total degree at most $2p$ and a constant reference metric, the Duffy map has degree at most $2p + n - 1$ in any individual cube coordinate after multiplication by its Jacobian. Consequently the tensor Gauss rule integrates such reference mass integrands exactly whenever $2q - 1 \geq 2p + n - 1$, that is, $q \geq p + \lceil n/2 \rceil$. Variable metrics produce nonpolynomial factors; in that case the quadrature estimate in Proposition 5.3 depends on their transformed derivatives. The rule $q \geq p + 4$ used in Sections 7 and 8 lies above this threshold for $n \leq 3$.

The standard FEEC theory makes the following passage from local compatible elements to stable Hodge problems [4, 5, 34]. It is useful to state exactly what the finite element implementation must supply.

Theorem 4.1 gives the subcomplex; stability and approximation further require a uniformly bounded cochain projection, which Proposition 4.5 and Theorem 4.6 supply for two mesh classes. The following assumption states this input for other mesh families.

Assumption 5.5 (Compatible intrinsic complex). The local form spaces, trace maps, and intrinsic chart family define a conforming subcomplex $V_{h,r}^\bullet \subset H\Lambda^\bullet(M)$ as in Theorem 4.1. There exist uniformly bounded projections $\pi_h^k : H\Lambda^k(M) \rightarrow V_{h,r}^k$ such that

$$\pi_h^{k+1}d = d\pi_h^k, \quad \pi_h^k|_{V_{h,r}^k} = I. \quad (5.5)$$

The triangulation admits the usual Whitney cochain map, with reference Whitney forms included in the local polynomial spaces and with compatible global facet traces.

Proposition 5.6 (Correct discrete cohomology under compatible projections). *Under Assumption 5.5, the inclusion $V_{h,r}^\bullet \hookrightarrow H\Lambda^\bullet(M)$ induces an isomorphism of cohomology. In particular, the number of discrete harmonic k -forms equals the Betti number $b_k(M)$.*

Proof. If a closed $v_h \in V_{h,r}^k$ is exact in the continuous complex, write $v_h = dw$. Applying (5.5) gives $v_h = d\pi_h^{k-1}w$, so the induced inclusion on cohomology is injective. For surjectivity, choose a simplicial cocycle representing any continuous de Rham class under the Whitney–de Rham

correspondence on the triangulated manifold. Its Whitney form lies in $V_{h,r}^k$ by assumption, and its image represents the prescribed class. Thus the inclusion is also surjective. Proposition 4.3 then identifies the harmonic dimension with $b_k(M)$. $\square$

Remark 5.7. Commuting projections on intrinsic smooth triangulations are constructed in [34, 35]. Proposition 5.6 applies to mesh families satisfying their uniform bounds and the facet compatibility condition (4.30).

Proposition 5.8 (Standard FEEC and spectral approximation result). *Assume 5.5, uniform approximation of the projection in the graph norm, the compactness properties required by the Hodge Hilbert complex, and metric-integral consistency. Then the mixed finite element Hodge–Laplace source problem is stable and convergent. If, in addition, the discrete solution operators converge in operator norm to the compact continuous solution operator on the orthogonal complement of harmonic forms, their isolated nonzero eigenvalues and spectral subspaces converge without spectral pollution.*

Proof. Assumption 5.5 supplies the two central FEEC hypotheses: a conforming subcomplex and uniformly bounded commuting projections. The abstract discrete Hodge decomposition and mixed inf–sup estimates therefore apply [4, 5]. Under the stated graph-norm approximation and compactness assumptions, the discrete solution operators converge as stated. The standard spectral approximation theorem for norm-convergent compact selfadjoint operators then yields convergence of the nonzero eigenvalues and their invariant subspaces [8, 13]. $\square$

The proposition uses the standard FEEC source approximation and compact selfadjoint operator spectral theory [4, 5, 8, 13].

Consider a second metric $\tilde{g}$ on the same conforming space, for example a piecewise constant approximation of the prescribed element tensor. The next theorem bounds the resulting spectral perturbation independently of the number of finite element degrees of freedom.

Theorem 5.9 (Uniform scalar metric-perturbation bound). *Let $0 < \varepsilon < 1$ and suppose the intrinsic metrics on a compact n -manifold satisfy, as quadratic forms on tangent vectors,*

$$(1 - \varepsilon)g \preceq \tilde{g} \preceq (1 + \varepsilon)g \quad \text{almost everywhere.} \quad (5.6)$$

Let $V_h \subset H^1(M)$ be any fixed finite-dimensional conforming space, with both scalar generalized eigenproblems assembled by exact integration on this same space. Number eigenvalues increasingly, counting multiplicity and retaining the constant zero mode if present. For each positive $\mu_{h,j}(g)$,

$$\frac{(1 - \varepsilon)^{(n-1)/2}}{(1 + \varepsilon)^{(n+1)/2}} \leq \frac{\mu_{h,j}(\tilde{g})}{\mu_{h,j}(g)} \leq \frac{(1 + \varepsilon)^{(n-1)/2}}{(1 - \varepsilon)^{(n+1)/2}}. \quad (5.7)$$

For each fixed manifold dimension n , the bounds are independent of $\dim V_h$ and of the finite element polynomial degree. The displayed factors depend explicitly on n . The dimension of the scalar kernel is unchanged.

Proof. For the volume densities, (5.6) gives $(1 - \varepsilon)^{n/2} dV_g \leq dV_{\tilde{g}} \leq (1 + \varepsilon)^{n/2} dV_g$. At each point diagonalize $\tilde{g}$ with respect to g ; write its generalized eigenvalues as $\lambda_i \in [1 - \varepsilon, 1 + \varepsilon]$. In the

corresponding g -orthonormal coframe, the stiffness density in coordinate direction i is multiplied by $\sqrt{\prod_{j \neq i} \lambda_j / \lambda_i}$. Hence

$$\frac{(1 - \varepsilon)^{(n-1)/2}}{(1 + \varepsilon)^{1/2}} a_g \leq a_{\tilde{g}} \leq \frac{(1 + \varepsilon)^{(n-1)/2}}{(1 - \varepsilon)^{1/2}} a_g.$$

The mass comparison follows from the product of the generalized eigenvalues. Divide the stiffness bounds by the opposite mass bounds and apply Courant–Fischer to obtain (5.7). Both Dirichlet forms vanish precisely on the locally constant functions; connectedness identifies the scalar kernel. $\square$

Theorem 5.10 (Fixed-space coefficient comparison). *Let ρ, G and $\tilde{\rho}, \tilde{G}$ be positive measurable scalar densities and positive-definite coefficient matrices on the same reference cells. Set $C = \rho G^{-1}$ and $\tilde{C} = \tilde{\rho} \tilde{G}^{-1}$. Suppose constants $0 < m_- \leq m_+$ and $0 < a_- \leq a_+$ satisfy, on all cells almost everywhere,*

$$m_- \rho \leq \tilde{\rho} \leq m_+ \rho, \quad a_- C \preceq \tilde{C} \preceq a_+ C.$$

Then every positive scalar eigenvalue computed on a fixed conforming space with exact integration satisfies

$$\frac{a_-}{m_+} \leq \frac{\mu_{h,j}(\tilde{C}, \tilde{\rho})}{\mu_{h,j}(C, \rho)} \leq \frac{a_+}{m_-}. \quad (5.8)$$

Proof. The assumptions give $m_- m(u, u) \leq \tilde{m}(u, u) \leq m_+ m(u, u)$ and $a_- a(u, u) \leq \tilde{a}(u, u) \leq a_+ a(u, u)$. Apply the min–max principle to the two generalized Rayleigh quotients. The comparison is algebraic on the fixed space, so the coefficient matrices may jump across cell facets. $\square$

For the three-dimensional conformal family $g = e^{2\phi} B$, $\tilde{g} = e^{2\tilde{\phi}} B$ with the same constant positive matrix B , write $t = \tilde{\phi} - \phi$. If $t_- \leq t(x) \leq t_+$ on every element, the mass and stiffness integrands are multiplied by e^{3t} and e^t , respectively; Theorem 5.10 gives

$$e^{t_- - 3t_+} \leq \frac{\mu_{h,j}(\tilde{g})}{\mu_{h,j}(g)} \leq e^{t_+ - 3t_-}. \quad (5.9)$$

This bound applies to the barycenter-frozen metric, whose values jump across facets.

Corollary 5.11 (Discretization and metric-data errors). *For every fixed positive continuum eigenvalue $\mu_j(g)$ and sufficiently small h , if the assumptions of Theorem 5.1 hold and (5.6) holds with $\varepsilon = \varepsilon_h$, then*

$$|\mu_{h,j}(\tilde{g}) - \mu_j(g)| \leq C_j h^{2p} + C_{j,n} \varepsilon_h \quad (5.10)$$

for fixed p , modulo the separately controlled quadrature error.

Proof. Apply the triangle inequality, (5.4), and Theorem 5.9. For ε_h bounded away from one, the ratios in (5.7) differ from one by $\mathcal{O}(\varepsilon_h)$. $\square$

If the approximate metric satisfies (5.6) with $\varepsilon_h = \mathcal{O}(h^q)$, the corresponding spectral perturbation is $\mathcal{O}(h^q)$. The rate q is determined by the regularity, chart family, and metric reconstruction. With the exact element metric, the remaining contributions in (5.10) are finite element approximation and quadrature errors.

The coordinate-transformation identity in Proposition 2.2 concerns two representations of the same global finite element space. Nonaffine reparametrization followed by fixed-degree polynomial interpolation generally produces different finite element spaces; their spectral errors are governed by their respective approximation properties.

Regularity of the metric and mesh.. The scalar h^{p+1} estimate in L^2 uses H^2 regularity of the adjoint problem, and the high-order energy estimate uses the regularity of the solution and of the charts through order $p+1$. For an interface metric with jumps, or a domain with corners or edges, the solution lies in H^{1+s} with $s < 1$, and the same arguments give the rates supported by that regularity [14, 17]. The trace relations of Assumption 2.1 describe matching facet triangulations, and the uniform atlas and projection bounds are hypotheses on the mesh family in the sense of [34].

5.1. Metric perturbations

For form problems the metric also enters the discrete adjoint of the exterior derivative, and the splitting into exact and coexact parts depends on it. We therefore compare spectra through the singular values of the fixed differential matrices.

Suppose g and $\tilde{g}$ satisfy (5.6). For each $0 \leq k \leq n$, define

$$a_k(\varepsilon) = \frac{(1-\varepsilon)^{(n-k)/2}}{(1+\varepsilon)^{k/2}}, \quad b_k(\varepsilon) = \frac{(1+\varepsilon)^{(n-k)/2}}{(1-\varepsilon)^{k/2}}. \quad (5.11)$$

The induced metric on the k th exterior power of the cotangent bundle obeys

$$(1+\varepsilon)^{-k} |\omega|_g^2 \leq |\omega|_{\tilde{g}}^2 \leq (1-\varepsilon)^{-k} |\omega|_g^2. \quad (5.12)$$

For sharper joint bounds, diagonalize $\tilde{g}$ against g . The integrand weight on the ordered component $dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ is multiplied by $\sqrt{\prod_{j \notin I} \lambda_j / \prod_{i \in I} \lambda_i}$. The resulting pointwise inequality, integrated on any fixed conforming form space, is

$$a_k(\varepsilon) M_k(g) \preceq M_k(\tilde{g}) \preceq b_k(\varepsilon) M_k(g). \quad (5.13)$$

These inequalities are independent of the number of form degrees of freedom and of the polynomial degree. They require that the two calculations use the same finite element complex and that the indicated mass matrices integrate the two metrics on that complex; changing the trace map or the trial space is a different perturbation problem.

Remark 5.12 (Relation to the Regge-metric analysis). For two Regge metrics Gawlik and McKee [23, Theorem 2] compare the L^2 norms of k -forms through the constants $C_{g_1, g_2} = \text{ess sup } \|v\|_{g_1} / \|v\|_{g_2}$ and the corresponding volume ratio. Under (5.6) one has $C_{\tilde{g}, g} \leq (1+\varepsilon)^{1/2}$ and $C_{g, \tilde{g}} \leq (1-\varepsilon)^{-1/2}$, and (5.13) is the matrix form of that comparison on the fixed complex, with the joint diagonalization supplying the factors (5.11). Their Theorem 17 bounds the geometric consistency operator of the Holst–Stern framework [24] by a multiple of $\|g - g_h\|_{L^\infty}$ and gives the source-problem error for a Regge metric g_h , given bounded commuting projections. Theorem 5.14 below treats the complementary quantity, the discrete Hodge spectrum. It orders every positive eigenvalue by an algebraic argument on the fixed complex, and it covers measurable cellwise tensors.

Let $D_j : V_h^j \rightarrow V_h^{j+1}$ be the fixed coefficient maps of a finite-dimensional complex with positive-definite inner products M_j . Write $\sigma_{j,1}, \dots, \sigma_{j,r_j} > 0$ for the nonzero singular values of D_j between these metric Hilbert spaces, where $r_j = \text{rank } D_j$. The nonzero eigenvalues of the discrete Hodge Laplacian in degree k are the multiset

$$\{\sigma_{k-1,i}^2 : 1 \leq i \leq r_{k-1}\} \cup \{\sigma_{k,i}^2 : 1 \leq i \leq r_k\}, \quad (5.14)$$

with missing endpoint terms suppressed. This is the finite-dimensional orthogonal Hodge decomposition: the first family belongs to the exact subspace, and the second to its metric-orthogonal complement within the nonharmonic sector. Although the decomposition depends on the metric, the ranks r_j do not.

Lemma 5.13 (Singular values under equivalent inner products). *Let $D : V \rightarrow W$ be a fixed finite-rank linear map. If two domain inner products and two range inner products satisfy*

$$a_V \|v\|_V^2 \leq \|v\|_{\tilde{V}}^2 \leq b_V \|v\|_V^2, \quad a_W \|w\|_W^2 \leq \|w\|_{\tilde{W}}^2 \leq b_W \|w\|_W^2,$$

then the ordered positive squared singular values satisfy

$$\frac{a_W}{b_V} \sigma_i(D)^2 \leq \tilde{\sigma}_i(D)^2 \leq \frac{b_W}{a_V} \sigma_i(D)^2. \quad (5.15)$$

Proof. The singular values of D are those of the induced injective map $\bar{D} : V/\ker D \rightarrow W$. The quotient norm is the minimum norm among representatives of a coset, so it inherits the same constants a_V, b_V . The squared singular values are the min-max values of $\|\bar{D}[v]\|_W^2 / \|[v]\|_{V/\ker D}^2$ on the finite-dimensional quotient. Comparing numerator and denominator and applying min-max proves the result. $\square$

Theorem 5.14 (Degree-independent perturbation bound for discrete Hodge spectra). *Let $V_h^\bullet$ be a fixed finite-dimensional differential complex, and suppose its Hodge matrices for the two metrics satisfy (5.13). Let*

$$\ell_k = \min_{j \in \{k-1, k\} \cap \{0, \dots, n-1\}} \frac{a_{j+1}}{b_j}, \quad u_k = \max_{j \in \{k-1, k\} \cap \{0, \dots, n-1\}} \frac{b_{j+1}}{a_j}, \quad (5.16)$$

where the factors are evaluated at ε . Number the positive discrete Hodge eigenvalues in degree k increasingly, with multiplicity. For every such eigenvalue,

$$\ell_k \mu_{h,i}^{(k)}(g) \leq \mu_{h,i}^{(k)}(\tilde{g}) \leq u_k \mu_{h,i}^{(k)}(g). \quad (5.17)$$

The number of zero eigenvalues is unchanged. The constants are independent of $\dim V_h^k$ and of the polynomial degree. For $k = 0$, the bounds reduce to the scalar estimate (5.7).

Proof. Apply Lemma 5.13 to each map D_j with $a_V = a_j$, $b_V = b_j$, $a_W = a_{j+1}$ and $b_W = b_{j+1}$. Every positive squared singular value of D_j lies between the corresponding lower and upper factors times its old value. The positive Hodge eigenvalues are obtained by merging the two ordered lists in (5.14). To see this for the merged lists, label the positive eigenvalues in both sectors before sorting and pair each value for g with its value for $\tilde{g}$ from the same differential map and the same index. Every paired value satisfies $\ell_k a_i \leq b_i \leq u_k a_i$. For any threshold t , at least j of the a_i lie below their j th order statistic; the corresponding b_i lie below u_k times that statistic. Hence the j th sorted b is at most u_k times the j th sorted a . Applying the same counting argument to the lower bounds gives the lower inequality. The pairing is made before merging and does not require the exact/coexact sector ordering to remain unchanged. Finally, the zero eigenvalue multiplicity equals $\dim \ker D_k - \text{rank } D_{k-1}$ by Proposition 4.3, which depends only on the fixed differential matrices. For $k = 0$, only $j = 0$ contributes and $a_1/b_0 = (1 - \varepsilon)^{n/2}/(1 + \varepsilon)^{n/2+1}$, with the analogous upper factor. $\square$

Remark 5.15 (Spectral clusters). The estimate concerns positive eigenvalues ordered with multiplicity. Eigenspace and spectral-projector bounds involve additional spectral gap conditions, particularly for multiple eigenvalues.

Suppose a mesh-dependent approximate metric $\tilde{g}_h$ satisfies

$$(1 - \varepsilon_h)g \preceq \tilde{g}_h \preceq (1 + \varepsilon_h)g, \quad \varepsilon_h \rightarrow 0. \quad (5.18)$$

For a fixed isolated eigenvalue cluster, the scalar estimate (5.10) separates the finite element contribution from $\mathcal{O}(\varepsilon_h)$. Theorem 5.14 gives the analogous metric-data comparison for each *discrete* positive Hodge eigenvalue on the same complex. Passing from these discrete comparisons to continuum Hodge eigenvalues additionally requires the source-operator convergence assumptions of Theorem 5.8.

For piecewise constant metric reconstruction in ordinary coordinates, $\varepsilon_h = \mathcal{O}(h)$ is a typical bound for a smooth reference metric. Higher-order reconstructions give the corresponding rate when the required regularity and approximation estimates hold. The metric perturbation term in (5.17) is controlled by the measured or estimated ε_h .

5.2. Boundary conditions and spectral approximation

Let M be closed for the moment and set $W^k = L^2\Lambda^k(M, g)$, $V^k = H\Lambda^k(M, g)$. Write δ for the L^2_g adjoint of d , with domain determined by the boundary condition when $\partial M \neq \emptyset$. The Hodge Laplacian is $\Delta_k = d\delta + \delta d$ and its harmonic space is

$$\mathcal{H}^k(M, g) = \{u \in V^k : du = 0, \delta u = 0\}. \quad (5.19)$$

For a given $f \in W^k$, a standard mixed weak formulation seeks $\sigma \in V^{k-1}$, $u \in V^k$, and $p \in \mathcal{H}^k$ such that

$$(\sigma, \tau)_g - (u, d\tau)_g = 0 \quad \forall \tau \in V^{k-1}, \quad (5.20)$$

$$(d\sigma, v)_g + (du, dv)_g + (p, v)_g = (f, v)_g \quad \forall v \in V^k, \quad (5.21)$$

$$(u, q)_g = 0 \quad \forall q \in \mathcal{H}^k. \quad (5.22)$$

The first equation represents $\sigma = \delta u$ weakly, and the final condition selects a unique solution on the orthogonal complement of the harmonic space. For $k = 0$ there is no σ term, and one recovers the scalar problem after removing constants or adding a positive reaction term.

Replacing V^k by the conforming spaces $V_{h,r}^k$ and $\mathcal{H}^k$ by the discrete harmonic space gives the mixed finite element problem. The discrete adjoint of the globally assembled derivative is

$$\delta_{h,k} = M_{k-1}^{-1} D_{k-1}^T M_k, \quad (5.23)$$

which produces the Hodge matrix (4.15). The source analysis uses the mixed formulation; the eigenvalue computations use the generalized eigenproblem for (4.15).

Assume a shape-regular family of intrinsic meshes, a uniformly bounded commuting projection as in (5.5), and the standard regularity and approximation conditions for the mixed problem. Let (σ, u, p) and (σ_h, u_h, p_h) solve the continuous and discrete mixed systems. Abstract FEEC error estimates bound their graph-norm error by best-approximation errors of σ, u, p , together with the

harmonic-space approximation term; see [3–5, 14, 33]. For smooth solutions and a fixed trimmed degree r , when the requisite local estimates hold, this yields the familiar h -order estimate

$$\|\sigma - \sigma_h\|_{H\Lambda^{k-1}} + \|u - u_h\|_{H\Lambda^k} \leq Ch^s \mathcal{N}_s(\sigma, u, p), \quad (5.24)$$

where s is the order supported simultaneously by the trimmed spaces, the solution regularity, and the chart family, and $\mathcal{N}_s$ collects the corresponding Sobolev norms. Under Assumption 5.5 the quasi-optimality of [5, Section 3] holds, and the trimmed space of order r approximates a form w with $w, dw \in H^r$ to order h^r in the graph norm. Hence $s = r$ when $\sigma, d\sigma, u, du$, and p lie in $H^r(M)$, which holds for the smooth data of Section 8.5; $O(h^r)$ is the order against which Tables 12 and 14 are read. An order h^{r+1} in L_g^2 for u requires in addition the elliptic regularity of the Hodge Laplacian used in the improved estimates of [5, Section 3].

Let $\partial M \neq \emptyset$. The tangential trace of a differential form is its pullback to ∂M , denoted $\text{tr}_{\partial M} u$. The relative complex is obtained by the essential condition

$$V_{\text{rel}}^k = \{u \in H\Lambda^k(M) : \text{tr}_{\partial M} u = 0\}. \quad (5.25)$$

The relation $\text{tr}_{\partial M} du = d_{\partial M} \text{tr}_{\partial M} u$ makes this a subcomplex. The discrete relative complex is obtained by imposing the same tangential-trace constraints on boundary entities. The complementary absolute boundary problem is defined using the L_g^2 adjoint domain. For smooth forms, absolute conditions can be written $\iota_\nu u = 0$ and $\iota_\nu du = 0$, while relative conditions can be written as the vanishing tangential trace together with the compatible adjoint boundary condition.

At degree zero, relative conditions are homogeneous Dirichlet and absolute conditions are homogeneous Neumann. An unpaired scalar boundary facet is therefore constrained or left free. At higher degrees the natural and essential conditions occur in different components of the mixed system, and the relative problem constrains the tangential trace coefficients alone.

The form computations in Section 8 use closed manifolds; the computations on manifolds with boundary are scalar Dirichlet and Neumann problems. Spectral convergence of the form problems follows from Theorem 5.8. For a multiple eigenvalue a finite mesh produces a cluster, and we report the cluster interval and its number of members.

6. Implementation

The constrained B-form formulation admits sparse extension, saddle-point, and augmented-Lagrangian realizations [7, 25]. We use the sparse extension matrix of Section 4.2; a dense nullspace of J_k on small meshes provides a regression test, and the two realizations are compared on the five-cell spherical mesh. The reference-element algebra follows [1, 6, 28].

The code has four parts: a metric provider, a reference-element module, a facet-constraint module, and an assembler. Each element stores its orientation, its ordered local entities, and a callable tensor $\hat{x} \mapsto G_K(\hat{x})$. Each identified facet stores its two incident cells, their local facet numbers, and the reference-facet transformation; an ordinary neighbor relation is a special case, and a facet without a partner receives a boundary trace condition. The metric interface is the same for an analytic model tensor, the pullback of a coefficient matrix, and a tensor supplied by a meshing program. Positivity of G_K is checked at the quadrature points, and the metric compatibility (2.6) of paired facets is checked at separate facet points.

6.1. Quadrature and matrix assembly

An n -simplex rule is a tensor-product Gauss rule with q points per direction on $[0, 1]^n$, mapped by the Duffy transformation [21]:

$$\hat{x}_j = t_j \prod_{m < j} (1 - t_m), \quad \left| \det \frac{\partial \hat{x}}{\partial t} \right| = \prod_{m=1}^{n-1} (1 - t_m)^{n-m}. \quad (6.1)$$

At each quadrature point with weight w the assembler evaluates G , its inverse and determinant, the basis values B_k and the reference gradients, and accumulates

$$M_{K,0} += w \sqrt{\det G} B_0^T B_0, \quad A_{K,0} += w \sqrt{\det G} (\nabla B_0)^T G^{-1} (\nabla B_0), \quad (6.2)$$

$$M_{K,k} += w \sqrt{\det G} B_k^T \mathcal{H}_k(G) B_k. \quad (6.3)$$

The element matrices are changed to the entity-moment basis by (4.21) and assembled by congruence with Z_k . Reference derivative and pullback matrices are reused for all facets with the same transformation, while the metric-dependent matrices are integrated cell by cell. The residuals (4.35), the harmonic dimensions, and the eigenvalue residuals are evaluated on the assembled operators. The error norms use a rule of higher order than the assembly, and the effect of the nonpolynomial metric factors is measured by increasing q .

7. Numerical experiments in two dimensions

7.1. Trefoil tube surface

We first consider a two-dimensional manifold with a non-diagonal intrinsic metric obtained from a tubular neighborhood of a trefoil knot. This experiment concerns the smooth *boundary surface* of the tube; the solid-torus problems of Section 8.3 concern its three-dimensional interior. The centerline is the standard (2,3)-trefoil

$$\gamma(t) = (\sin t + 2 \sin(2t), \cos t - 2 \cos(2t), -\sin(3t)), \quad t \in \mathbb{R}/2\pi\mathbb{Z}. \quad (7.1)$$

Let $\ell(t) = |\gamma'(t)|$ and let $\kappa(t)$ and $\tau(t)$ be its curvature and torsion. With the Frenet normal and binormal $n(t)$ and $b(t)$, respectively, the tube surface of radius $a = 0.22$ has the periodic parametrization

$$F(t, \vartheta) = \gamma(t) + a(\cos \vartheta n(t) + \sin \vartheta b(t)), \quad (t, \vartheta) \in (\mathbb{R}/2\pi\mathbb{Z})^2. \quad (7.2)$$

Its metric is supplied to the finite element method by the following intrinsic coordinate matrix:

$$G(t, \vartheta) = \begin{pmatrix} \ell^2 [(1 - a\kappa \cos \vartheta)^2 + a^2 \tau^2] & \ell a^2 \tau \\ \ell a^2 \tau & a^2 \end{pmatrix}, \quad \rho(t, \vartheta) = \sqrt{\det G} = a \ell(t) (1 - a\kappa(t) \cos \vartheta). \quad (7.3)$$

Here ℓ , κ , and τ are evaluated at t . The maximum sampled value of $a\kappa$ is 0.17090. The coordinate metric therefore remains positive definite throughout the parameter square. The length integral of the centerline gives the independent surface-area reference

$$|M|_g = 2\pi a \int_0^{2\pi} \ell(t) dt = 39.84660306847. \quad (7.4)$$

The surface is a torus as an abstract manifold, while its prescribed metric contains the varying speed, curvature, and torsion of the knotted centerline. The off-diagonal coefficient in (7.3) is included in every element integral. In the finite element calculation, the parametrization (7.2) is used to prescribe the metric; the element assembler operates on (7.3) and the two periodic facet identifications.

For the manufactured Poisson problem we prescribe

$$w(t, \vartheta) = \sin(2t + \vartheta) + 0.35 \cos(3t - 2\vartheta), \quad f = -\rho^{-1} \partial_i (\rho (G^{-1})^{ij} \partial_j w), \quad (7.5)$$

with indices ranging over t, ϑ . The discrete solution has zero integral, and the L_g^2 error is measured modulo constants. We use $S_p^0(\Delta)$ on a doubly periodic mesh with $2N_t N_\vartheta$ triangles and $N_t N_\vartheta p^2$ coefficients. The centerline has length 28.826 and the cross-section has circumference 1.382, so the meshes are elongated along t and are refined by doubling in both directions, so the observed order is $\log_2$ of the ratio of successive errors. Table 1 shows the orders $p + 1$ in L_g^2 and p in the H_g^1 seminorm for degrees two through six.

N_t	N_ϑ	p	DOFs	$E_{L_g^2}$	order	$E_{H_g^1}$	order
16	4	2	256	$2.836 \cdot 10^{-1}$	—	$4.361 \cdot 10^0$	—
32	8	2	1024	$4.064 \cdot 10^{-2}$	2.80	$1.177 \cdot 10^0$	1.89
64	16	2	4096	$4.809 \cdot 10^{-3}$	3.08	$3.059 \cdot 10^{-1}$	1.94
16	4	3	576	$4.000 \cdot 10^{-2}$	—	$8.561 \cdot 10^{-1}$	—
32	8	3	2304	$2.618 \cdot 10^{-3}$	3.93	$1.148 \cdot 10^{-1}$	2.90
64	16	3	9216	$1.487 \cdot 10^{-4}$	4.14	$1.459 \cdot 10^{-2}$	2.98
16	4	4	1024	$4.586 \cdot 10^{-3}$	—	$1.424 \cdot 10^{-1}$	—
32	8	4	4096	$1.456 \cdot 10^{-4}$	4.98	$9.291 \cdot 10^{-3}$	3.94
64	16	4	16384	$4.428 \cdot 10^{-6}$	5.04	$5.901 \cdot 10^{-4}$	3.98
16	4	5	1600	$4.577 \cdot 10^{-4}$	—	$1.882 \cdot 10^{-2}$	—
32	8	5	6400	$7.373 \cdot 10^{-6}$	5.96	$6.297 \cdot 10^{-4}$	4.90
64	16	5	25600	$1.139 \cdot 10^{-7}$	6.02	$1.992 \cdot 10^{-5}$	4.98
16	4	6	2304	$4.477 \cdot 10^{-5}$	—	$2.313 \cdot 10^{-3}$	—
32	8	6	9216	$3.556 \cdot 10^{-7}$	6.98	$3.709 \cdot 10^{-5}$	5.96
64	16	6	36864	$2.793 \cdot 10^{-9}$	6.99	$5.864 \cdot 10^{-7}$	5.98

Table 1: Manufactured Poisson problem on the trefoil tube surface with the metric (7.3). Gauss–Duffy rules have $\max(8, p + 4)$ points in each direction for assembly and $\max(10, p + 6)$ for the error norms. The H_g^1 column is the error seminorm.

The same matrices give the eigenproblem $K_{hc} = \lambda_h M_{hc}$, with eigenvalues indexed from $\lambda_{h,0} = 0$. Table 2 reports values through index 480 at $p = 5$. An eigenfunction of index 480 has about 33 wavelengths along the knot and two around the tube, so the resolution is needed in t . The sequence refined along t converges to $\lambda_{480} = 154.30$, with successive changes 1.35, 0.070, and 0.012. The leading Weyl term $4\pi \cdot 480 / |M|_g = 151.4$ with the area (7.4) gives the scale. The last row uses a mesh refined around the tube; at a comparable number of unknowns its value at index 480 is 190.81, which shows the gain from placing the resolution along the knot. The two lowest positive eigenvalues coincide to fourteen digits, as the dihedral symmetry of the knot requires.

Figure 4 shows eight eigenfunctions from the finest run. The eigenfunctions below $1/a^2 = 20.66$ vary only along the knot. Index 43 is the first with one oscillation around the tube, and the branch with two oscillations opens near $4/a^2 = 82.64$. At high axial wavenumber the eigenfunctions concentrate near $\vartheta = \pi$, where the factor $1 - \kappa \cos \vartheta$ in the metric is largest; index 470 has amplitude about 0.15 of its maximum along $\vartheta = 0$.

The area computed from the element quadrature agrees with (7.4) within 10^{-11} , and the

N_t	N_ϑ	DOFs	$\lambda_{h,1}$	$\lambda_{h,80}$	$\lambda_{h,120}$	$\lambda_{h,240}$	$\lambda_{h,480}$
48	8	9600	0.047820712	24.338113	33.848370	83.037561	155.731512
72	8	14400	0.047820712	24.338110	33.848340	83.037561	154.380985
96	8	19200	0.047820712	24.338110	33.848339	83.037561	154.310866
144	12	43200	0.047820712	24.338110	33.848338	83.037561	154.298769
24	18	10800	0.047820712	24.338402	34.090263	84.262874	190.806508

Table 2: Laplace–Beltrami eigenvalues on the trefoil tube surface; $p = 5$, metric (7.3), order-twelve Gauss–Duffy quadrature, eigenvalues indexed from zero. The relative algebraic residuals of the listed eigenpairs are below $2 \cdot 10^{-11}$.

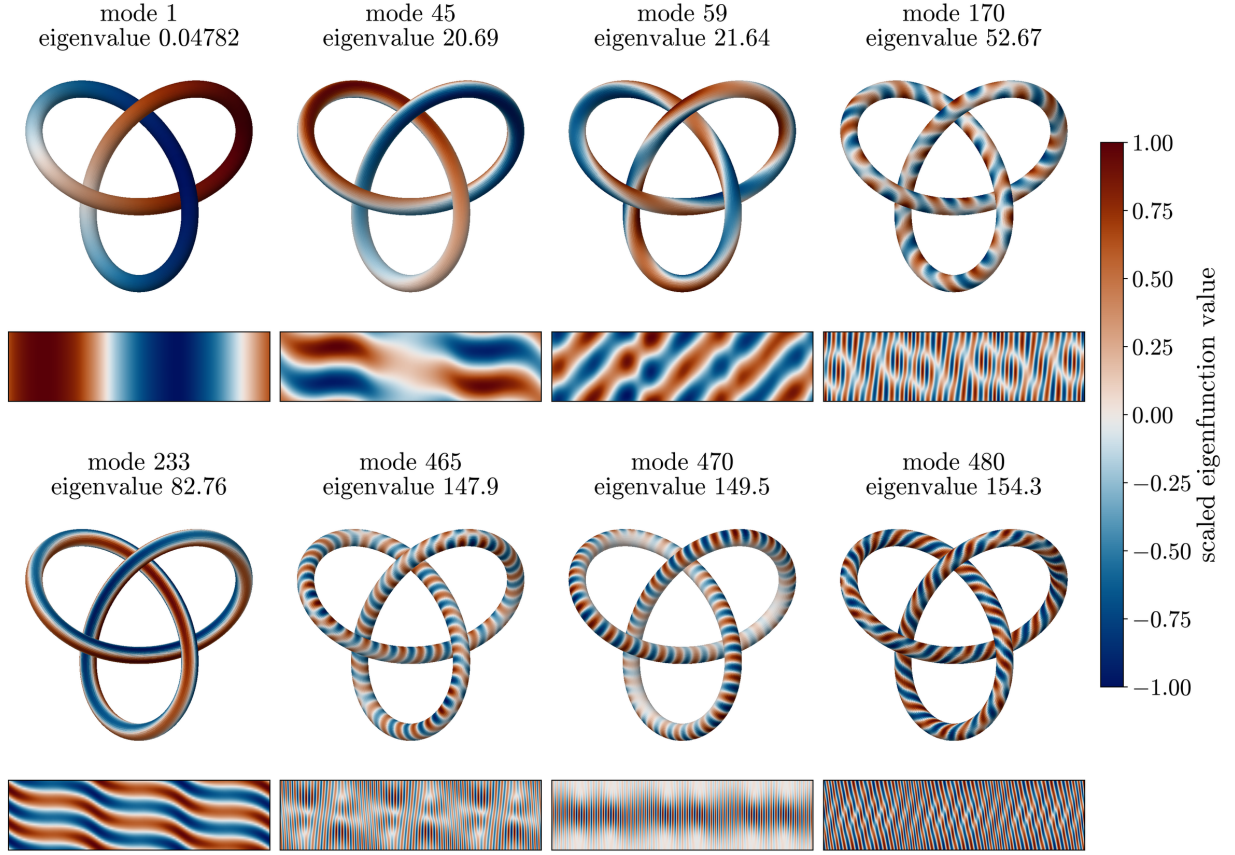


Figure 4: Laplace–Beltrami eigenfunctions on the trefoil tube surface, $p = 5$ on the 144×12 mesh. Each function is drawn on the embedded surface and in the parameter rectangle with t horizontal and ϑ vertical.

manufactured source agrees with an independently differentiated flux to within $6 \cdot 10^{-9}$ at three parameter points.

8. Numerical experiments in three dimensions

We now consider three-dimensional intrinsic finite element problems. We report scalar and differential-form approximation errors, discrete-complex residuals, and computed spectra, using the

element metrics of the corresponding manifolds.

8.1. Variable-metric torus

Let $M = \mathbb{R}^3/\mathbb{Z}^3$ have periodic coordinates $x = (x_1, x_2, x_3)$. We write the metric in the standard coordinate form

$$g = \sum_{i,j=1}^3 g_{ij}(x) dx_i \otimes dx_j, \quad g_{ij}(x) = e^{2\phi(x)} B_{ij}, \quad (8.1)$$

where

$$\phi(x) = 0.12 \prod_{i=1}^3 \sin(2\pi x_i) + 0.08 \cos(2\pi(x_1 + x_2)). \quad (8.2)$$

For the principal run $B = I_3$; a second run uses

$$B = \begin{pmatrix} 1.20 & 0.17 & 0.04 \\ 0.17 & 0.91 & 0.08 \\ 0.04 & 0.08 & 1.10 \end{pmatrix}. \quad (8.3)$$

The coefficient matrix $(g_{ij}) = e^{2\phi} B$ is positive definite and periodic. Each tetrahedral cell with affine coordinate map $x = x_K + E_K \hat{x}$ has the reference metric

$$G_K(\hat{x}) = e^{2\phi(x_K + E_K \hat{x})} E_K^T B E_K. \quad (8.4)$$

The exact solution is

$$u(x) = \sin(2\pi x_1) + 0.35 \cos(2\pi x_2) + 0.20 \sin(2\pi(x_1 + x_2 + x_3)), \quad (8.5)$$

The forcing is calculated in the same metric as the weak problem. In coordinates on $\mathbb{R}^3/\mathbb{Z}^3$, for $g = e^{2\phi} B$ with B constant positive definite, the scalar Laplace–Beltrami operator satisfies

$$\Delta_g u = e^{-2\phi} [\text{tr}(B^{-1} \nabla^2 u) + (\nabla \phi)^T B^{-1} \nabla u]. \quad (8.6)$$

This follows by inserting $g^{ij} = e^{-2\phi} (B^{-1})_{ij}$ and $\sqrt{\det g} = e^{3\phi} \sqrt{\det B}$ into (3.5). The forcing for (8.5) is

$$f = u - e^{-2\phi} [\text{tr}(B^{-1} \nabla^2 u) + (\nabla \phi)^T B^{-1} \nabla u]. \quad (8.7)$$

All coefficients are periodic. The exact solution and its coordinate gradient are evaluated using quadrature independent of the solution assembly. The L_g^2 and H_g^1 errors use respectively ρ_K and $G_K^{-1} \rho_K$ in the reference-element integrals.

The forcing $f = (-\Delta_g + I)u$ is evaluated from (8.7). We use continuous Bernstein polynomials with periodic coefficient identification and report the intrinsic L^2 and H^1 seminorm errors. The tabulated N is the number of periodic grid intervals in each coordinate direction; there are $6N^3$ tetrahedra.

For an error E_N at a fixed polynomial degree and two mesh sizes $N_- < N$, we report the observed order

$$s_E(N_-, N) = \frac{\log(E_{N_-}/E_N)}{\log(N/N_-)}, \quad h_N \simeq N^{-1}. \quad (8.8)$$

A dash marks the first mesh of each fixed-degree sequence.

N	p	DOFs	E_{L^2}	s_{L^2}	E_{H^1}	s_{H^1}
4	1	64	2.0860×10^{-1}	—	2.5377×10^0	—
6	1	216	1.2035×10^{-1}	1.36	1.8790×10^0	0.74
8	1	512	7.6629×10^{-2}	1.57	1.4751×10^0	0.84
12	1	1728	3.7587×10^{-2}	1.76	1.0192×10^0	0.91
4	2	512	3.7825×10^{-2}	—	8.3159×10^{-1}	—
6	2	1728	1.1415×10^{-2}	2.95	4.2169×10^{-1}	1.67
8	2	4096	4.8179×10^{-3}	3.00	2.5342×10^{-1}	1.77
4	3	1728	6.8983×10^{-3}	—	2.4270×10^{-1}	—
6	3	5832	1.4154×10^{-3}	3.91	7.9131×10^{-2}	2.76
8	3	13824	4.3370×10^{-4}	4.11	3.4016×10^{-2}	2.93
3	4	1728	4.5649×10^{-3}	—	1.5331×10^{-1}	—
4	4	4096	1.1171×10^{-3}	4.89	5.3499×10^{-2}	3.66
5	4	8000	3.7566×10^{-4}	4.88	2.3021×10^{-2}	3.78
6	4	13824	1.5571×10^{-4}	4.83	1.1463×10^{-2}	3.82
4	5	8000	1.9697×10^{-4}	—	1.0626×10^{-2}	—
5	5	15625	5.4678×10^{-5}	5.74	3.6690×10^{-3}	4.77

Table 3: Manufactured scalar solution on the three-torus with $g_{ij} = e^{2\phi}\delta_{ij}$ and ϕ as in (8.2). The observed orders compare successive displayed values of N at fixed degree; s_{L^2} and s_{H^1} are defined by (8.8).

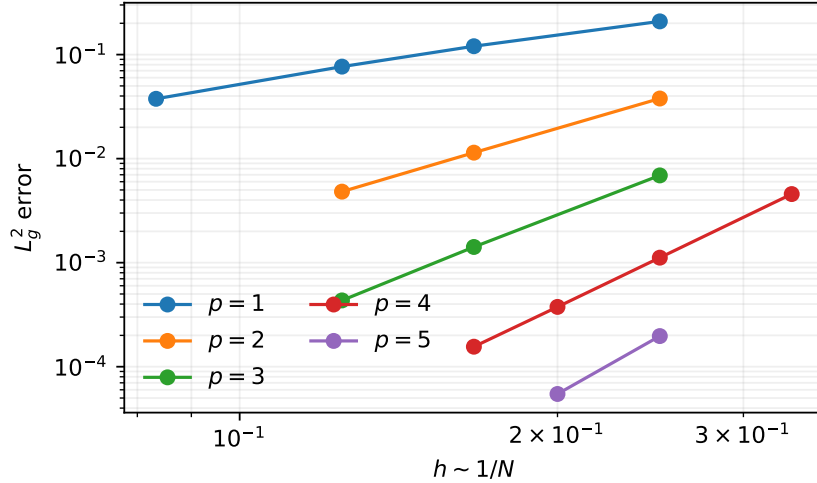
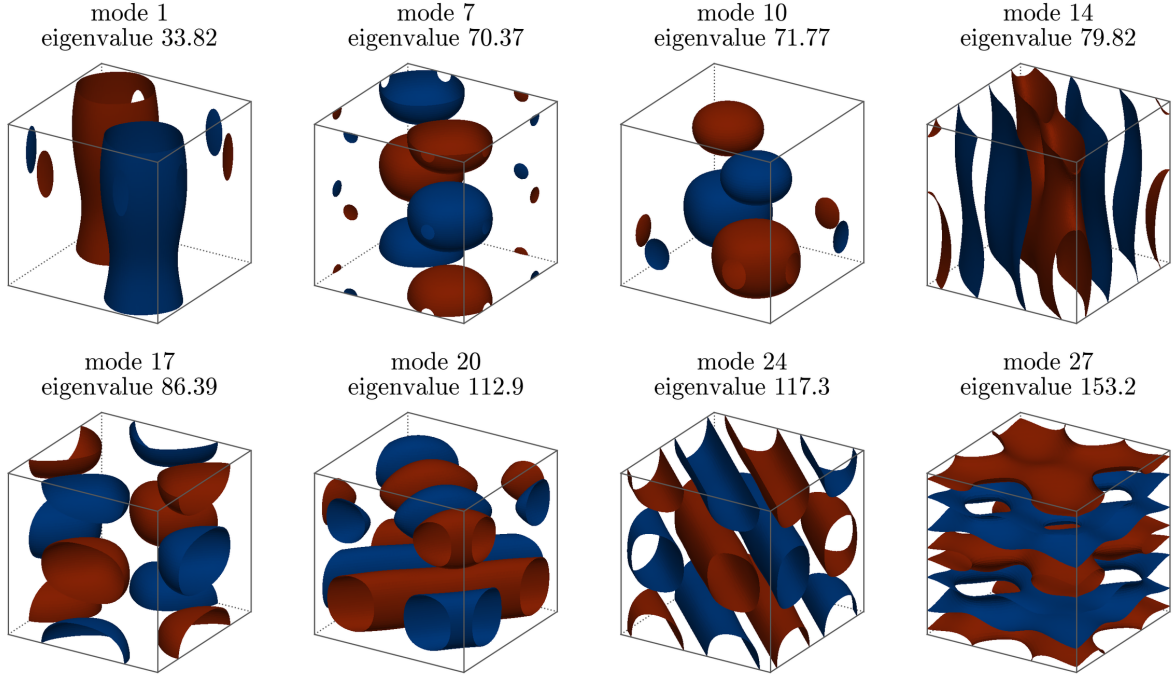


Figure 5: Absolute scalar L_g^2 errors on the periodic three-torus for fixed degrees $p = 1, \dots, 5$. The abscissa is the coordinate mesh scale $1/N$; the data are those of Table 3. The degree-four assembly used two quadrature orders, as stated in Section 8.4.

Table 3 and Figure 5 record the errors. For $p \geq 3$ the observed orders reach $p + 1$ in L_g^2 and p in the H_g^1 seminorm, as in Theorem 5.1; at $p = 1, 2$ the H_g^1 orders on the finest pairs are 0.91 and 1.77 and are still increasing. Each cube of the periodic grid is divided into six tetrahedra by the Kuhn (Freudenthal) subdivision, and $h_N = \sqrt{3}/N$ is the cube diagonal. The runs with the non-diagonal tensor (8.3) are in Table 8.

The same stiffness and mass matrices give the Laplace–Beltrami eigenproblem. Figure 6 shows eight eigenfunctions computed with $N = 6$ and $p = 4$ (13,824 unknowns). The first positive eigenvalue is 33.816, in agreement with the refinement sequence in Table 16. On the flat torus the first positive level $4\pi^2$ has multiplicity six. The conformal factor (8.2) resolves it into six simple eigenvalues 33.82, 34.73, 37.72, 38.62, 43.12, 44.08, and the tenth eigenfunction concentrates in one part of the periodic cell.



surfaces where the scaled eigenfunction equals 0.45 (brown) and minus 0.45 (blue)

Figure 6: Laplace–Beltrami eigenfunctions on the three-torus with the metric $g_{ij} = e^{2\phi}\delta_{ij}$ and ϕ as in (8.2); $N = 6$, $p = 4$. Each eigenfunction is scaled to unit maximum modulus, and the two surfaces are its level sets at 0.45 and -0.45 inside one periodic cell.

On the same periodic torus with $N = 2$ and $p = r = 2$, we assembled the scalar Laplacian in two ways: directly from the Bernstein scalar mass and stiffness matrices, and by the compatible complex through $D_0^T M_1 D_0$ and the scalar form mass matrix. The relative discrepancy between their *full generalized scalar eigenvalue lists* was $2.00 \cdot 10^{-15}$ for $B = I$ and $2.19 \cdot 10^{-15}$ for (8.3). This agreement checks metric integration, global constraints, and the first discrete differential.

The periodic mesh has 48 tetrahedra. The assembled form spaces have dimensions $(8, 56, 96, 48)$ at $r = 1$ and $(64, 304, 432, 192)$ at $r = 2$; both complexes have cohomology dimensions $(1, 3, 3, 1)$, and the one-form Hodge matrix at $r = 2$ has three zero modes.

8.2. Spherical three-manifold

Let $M = S^3/\Gamma$, where S^3 has its unit-round metric g_{S^3} (constant sectional curvature $+1$) and Γ is the binary icosahedral group of order 120 acting freely by isometries. The metric g on the quotient is determined by

$$\pi^*g = g_{S^3}, \quad \pi : S^3 \longrightarrow S^3/\Gamma. \quad (8.9)$$

The spherical space-form representation follows standard quotient constructions [30]. Group-invariant finite elements and numerical Laplace–Beltrami spectra on this manifold were previously computed by Bachelot-Motet [9]; the experiment here tests the intrinsic metric and the constrained differential complex on the five-cell quotient and its refinement. The computational mesh contains five geodesic tetrahedra from the 600-cell quotient. The element chart used in the computation can

be specified explicitly. Let $v_0, \dots, v_3 \in S^3$ be a cell's vertices in the standard unit-sphere model, set $y(\hat{x}) = v_0 + E_K \hat{x}$ and $E_K = [v_1 - v_0, v_2 - v_0, v_3 - v_0]$, and use the radial chart $F_K(\hat{x}) = y(\hat{x})/|y(\hat{x})|$ followed by the quotient map. The reference-cell metric matrix is

$$G_K(\hat{x}) = E_K^T \left(\frac{I_4}{|y(\hat{x})|^2} - \frac{y(\hat{x})y(\hat{x})^T}{|y(\hat{x})|^4} \right) E_K. \quad (8.10)$$

This matrix represents the pullback of the intrinsic quotient metric in (8.9); the finite element integrals use only its three-by-three entries. A symmetric refinement gives forty tetrahedra. The paired facets include nontrivial vertex permutations.

The dimensions of the assembled spaces and the composition residuals are in Table 10; every assembled complex has harmonic dimensions $(1, 0, 0, 1)$. In the forty-cell run at $r = 3$, the three lowest computed coexact one-form eigenvalues lie near 4.000004. They approximate the three left-invariant one-forms descending from the round three-sphere, whose exact coexact Hodge eigenvalue is 4 with multiplicity three.

mesh	order	scalar DOFs	$\mu_{h,1}$	$ \mu_{h,1} - 168 $
5 tetrahedra	2	7	257.795806	89.795806
5 tetrahedra	3	23	180.586	12.586
40 tetrahedra	2	54	172.812033	4.812033
40 tetrahedra	3	181	168.371939	0.371939
continuum reference			168	0

Table 4: First positive scalar eigenvalue on S^3/Γ with the unit-round quotient metric (8.9); the continuum reference is obtained by restricting round-sphere harmonics to the binary-icosahedral invariant subspace [30]. At $r = 3$ on the forty-cell mesh, the computed thirteen-eigenvalue cluster lies in $[168.37194, 171.36661]$, with width 2.99467. The continuum first positive level is 168 with multiplicity 13 in the invariant spherical harmonic spectrum [30]; the table reports the full computed cluster.

The integrated quotient volume on the five-cell mesh differs from $2\pi^2/120$ by $6.3 \cdot 10^{-15}$. At fixed order, refinement from five to forty tetrahedra reduces the first-eigenvalue error in Table 4 by factors of 18.7 at $r = 2$ and 33.8 at $r = 3$.

The scalar space $S_p^0(\Delta)$ on the five cells reaches much higher accuracy through the degree. At $p = 14$ the space has 2289 unknowns, and Table 5 lists the first computed levels. The continuum levels are $k(k+2)$ with multiplicity $k+1$ for $k = 12, 20, 24, 30, 32$ [30]; each computed cluster has the correct number of members. The mesh of this computation is explicit. With the unit quaternions $v_0 = -1$, $v_{1,2} = (-c, -\frac{1}{2}, \mp s, 0)$, $v_{3,4} = (-c, -s, 0, \mp \frac{1}{2})$, $v_{5,6} = (-c, 0, -\frac{1}{2}, \mp s)$, where $c = \cos(\pi/5)$ and $s = \cos(2\pi/5)$, the five tetrahedra of the 600-cell around the edge $v_0 v_1$,

$$K_0 = [v_0 v_1 v_2 v_3], \quad K_1 = [v_0 v_1 v_2 v_4], \quad K_2 = [v_0 v_1 v_3 v_5], \quad K_3 = [v_0 v_1 v_4 v_6], \quad K_4 = [v_0 v_1 v_5 v_6],$$

contain one tetrahedron from each orbit of Γ acting by left multiplication. Table 6 lists the ten facet pairings. Five are the interior faces around the edge, and five are induced by group elements; the degrees of freedom are identified through the orbits of the domain points. The clusters at 960 and 1088 are wider because the corresponding harmonic polynomials have degrees 30 and 32 and are less well resolved by degree 14 on five cells.

Figure 7 shows eigenfunctions from these levels on the dodecahedral fundamental cell, the Voronoi cell of one vertex of the 600-cell, in stereographic projection. Conjugation by the elements

level	multiplicity	smallest computed value	largest computed value
168	13	167.9999999999	168.0000000003
440	21	440.0000005	440.0000023
624	25	624.000004	624.000041
960	31	960.0013	960.0164
1088	33	1088.0095	1088.0276

Table 5: Scalar eigenvalue clusters on S^3/Γ computed with $S_{14}^0(\Delta)$ on the five-cell mesh (2289 unknowns) and the metric (8.10).

of Γ acts on the quotient by isometries that rotate this cell, and we project each eigenspace onto the functions invariant under a chosen rotation subgroup. The full icosahedral rotation group leaves one function invariant in each of the levels 168, 440, 624, and 960, in agreement with the decomposition of the corresponding representations of $SO(3)$; the measured projectors have rank one. The second row shows functions invariant under the rotations about a five-fold and a three-fold axis.

cell	facet	cell	facet	vertex map	group element γ
0	0	4	1	(2 0 1)	$(c, -\frac{1}{2}, s, 0)$
0	1	3	0	(2 1 0)	$(c, 0, \frac{1}{2}, -s)$
1	0	2	1	(2 1 0)	$(c, -s, 0, \frac{1}{2})$
1	1	4	0	(1 0 2)	$(c, 0, \frac{1}{2}, s)$
2	0	3	1	(1 0 2)	$(c, -s, 0, -\frac{1}{2})$
0	2	2	3	(0 1 2)	1
0	3	1	3	(0 1 2)	1
1	2	3	3	(0 1 2)	1
2	2	4	3	(0 1 2)	1
3	2	4	2	(0 1 2)	1

Table 6: Facet pairings of the five-cell mesh of S^3/Γ . Facet f of a cell is opposite its vertex f . The element γ carries the first facet onto the second by left multiplication, and the vertex map gives, for each vertex of the first facet in increasing order, the position of its image in the second.

8.3. Volume eigenvalue problems

The scalar volume experiments use the periodic parameter domain $M = (\mathbb{R}/2\pi\mathbb{Z}) \times D_{a,h}$, where $D_{a,h} \subset \mathbb{R}^2$ is the polygonal disk obtained by triangulating the cross-section of radius $a = 0.35$ in Cartesian coordinates (u, v) . The axial coordinate is $s \in \mathbb{R}/2\pi\mathbb{Z}$. The element map from $\hat{K}$ to the coordinates (s, u, v) is affine, and its pullback of the following coordinate metric gives G_K in (2.1).

For the standard solid torus of major radius $R = 2.2$, the metric is

$$g_{\text{tor}} = (R + u)^2 ds^2 + du^2 + dv^2. \quad (8.11)$$

The twisted-coordinate test uses the same physical metric in a second periodic chart. With $m = 1$ and

$$\begin{aligned} u' &= u \cos(ms) - v \sin(ms), \\ v' &= u \sin(ms) + v \cos(ms), \end{aligned} \quad g_{\text{tw}} = (R + u')^2 ds^2 + (du - mv ds)^2 + (dv + mu ds)^2. \quad (8.12)$$

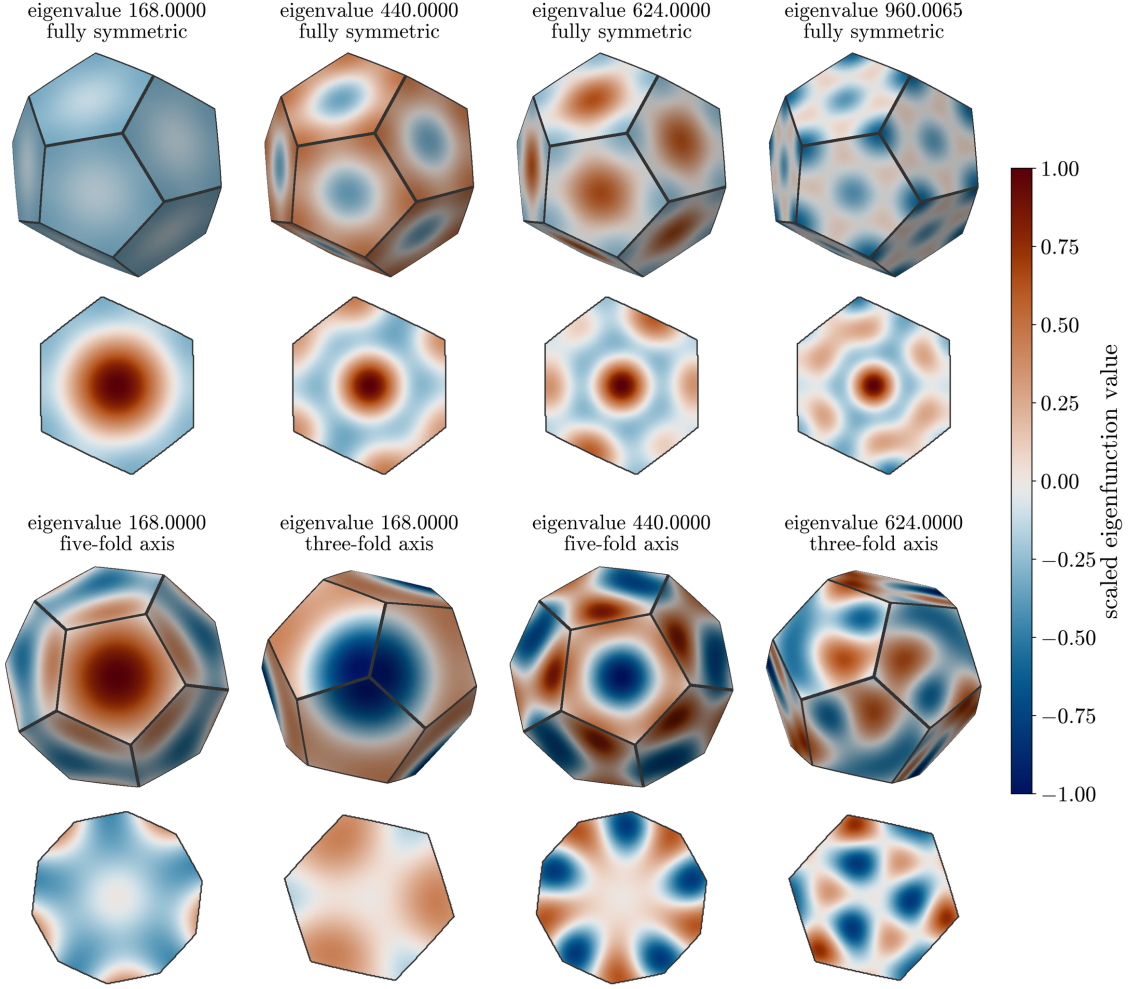


Figure 7: Eigenfunctions on the Poincaré dodecahedral space S^3/Γ , computed with $S_{14}^0(\Delta)$ on five cells. Each function is drawn on the boundary of the dodecahedral fundamental cell and, below it, on the plane section through the center of the cell orthogonal to the viewing direction. Opposite faces carry the same values after the identifying rotation. First row: the function invariant under all rotations of the cell in each level. Second row: functions invariant under the rotations about one axis.

The two metrics are related by the displayed coordinate transformation. On a circular disk D_a , they describe the same Riemannian solid torus. The standard and twisted parametrizations are related nonlinearly, so they give two different fixed-degree discrete spaces for one Riemannian problem, and Table 7 compares the two approximations.

The polygonal disks $D_{a,h}$ yield two distinct finite-domain approximations of that circular model. The third test uses a tube around the closed centerline

$$c(s) = (\sin s + 2 \sin(2s), \cos s - 2 \cos(2s), -\sin(3s)). \quad (8.13)$$

Write $q(s) = |c'(s)|$, $\kappa(s) = |c'(s) \times c''(s)|/q(s)^3$, and $\tau(s) = \det(c'(s), c''(s), c^{(3)}(s))/|c'(s) \times c''(s)|^2$.

In the moving-frame coordinates (s, u, v) used by the metric provider, its metric is

$$g_{\text{tr}} = q^2(1 - \kappa u)^2 ds^2 + (du - q\tau v ds)^2 + (dv + q\tau u ds)^2. \quad (8.14)$$

This tensor differs from (8.11); in particular, the speed, curvature, and torsion of the centerline enter the volume operator. All three problems use tetrahedral *volume* elements, periodicity in s , and either zero normal flux or zero scalar trace on the lateral boundary.

Table 7 lists the first positive Neumann eigenvalues. For the round tube the thin-tube value $R^{-2} = 0.20661$ gives the scale of the quadratic result 0.20786. The first Dirichlet eigenvalues on the 3456-cell linear mesh are 49.67230 and 49.68246 for the standard and twisted charts. The two charts give different fixed-degree trial spaces, as in statement (iii) of Section 2.1; the agreement of the leading digits reflects the convergence of two different discrete problems to one continuum problem.

metric/chart	tetrahedra	polynomial degree	first positive Neumann eigenvalue
round torus	1296	1	0.2126307
reparametrized torus	1296	1	0.2126311
round torus	3456	1	0.2090555
reparametrized torus	3456	1	0.2090556
round torus	1296	2	0.2078609
reparametrized torus	1296	2	0.2078608
changed-centerline tube	3456	1	0.0482809

Table 7: Scalar volume eigenvalues on the specified periodic tetrahedral meshes with polygonal cross-sectional disks, for g_{tor} , g_{tw} , and g_{tr} in (8.11)–(8.14). The values are eigenvalues of the polygonal computational domains. The coordinate twist reparametrizes the standard torus; the changed centerline gives a different tube metric.

Figure 8 shows eigenfunctions of the solid trefoil tube computed with $p = 3$ on 96 axial layers and a cross-section of 54 triangles (15,552 tetrahedra, 78,048 unknowns). The first positive Neumann eigenvalue is 0.0479. The Neumann eigenvalues below 28 belong to functions that are nearly constant on each cross-section. The first transverse branch begins at 28.25, near $(j'_{1,1}/a)^2 = 27.7$ for the disk of radius a . In that branch the nodal line of the cross-section rotates along the centerline in the Frenet coordinates (u, v) , which is the effect of the torsion terms in (8.14). The Dirichlet ground state, 48.16 against $(j_{0,1}/a)^2 = 47.2$ for the straight tube, has three maxima along the knot, on the three arcs where the curvature is largest; its amplitude on the centerline varies by 20 percent and has correlation 0.88 with κ .

8.4. Quadrature orders, anisotropic refinement, and reference conditioning

The scalar torus assembly uses $q = 6$ Gauss points per Duffy coordinate at $p \leq 2$, $q = 7$ at $p = 3$, $q = 8$ or 9 at $p = 4$, and $q = 10$ at $p = 5$; the error norms use a separate rule of higher order. The one-form integration orders appear in Table 14. Table 8 gives the runs with the non-diagonal tensor (8.3), in which the full inverse metric enters every element integral.

Table 9 gives the condition numbers of the reference moment matrices of the trimmed family on $\hat{K}$. These matrices belong to the reference element alone. The Bernstein change of basis preserves the space and lowers the condition numbers by about two orders of magnitude.

Table 11 reports the chart regularity of the meshes in use, in the sense of Assumption 2.1 and Theorem 5.1. The variation is the range of the generalized eigenvalues of $G_K(\hat{x})$ against G_K at the barycenter over a grid of points in each cell. The anisotropy is $\lambda_{\max}/\lambda_{\min}$ of G_K at the barycenter; it measures the shape of the cell against the right-angled reference simplex, and equals 4 for a regular tetrahedron.

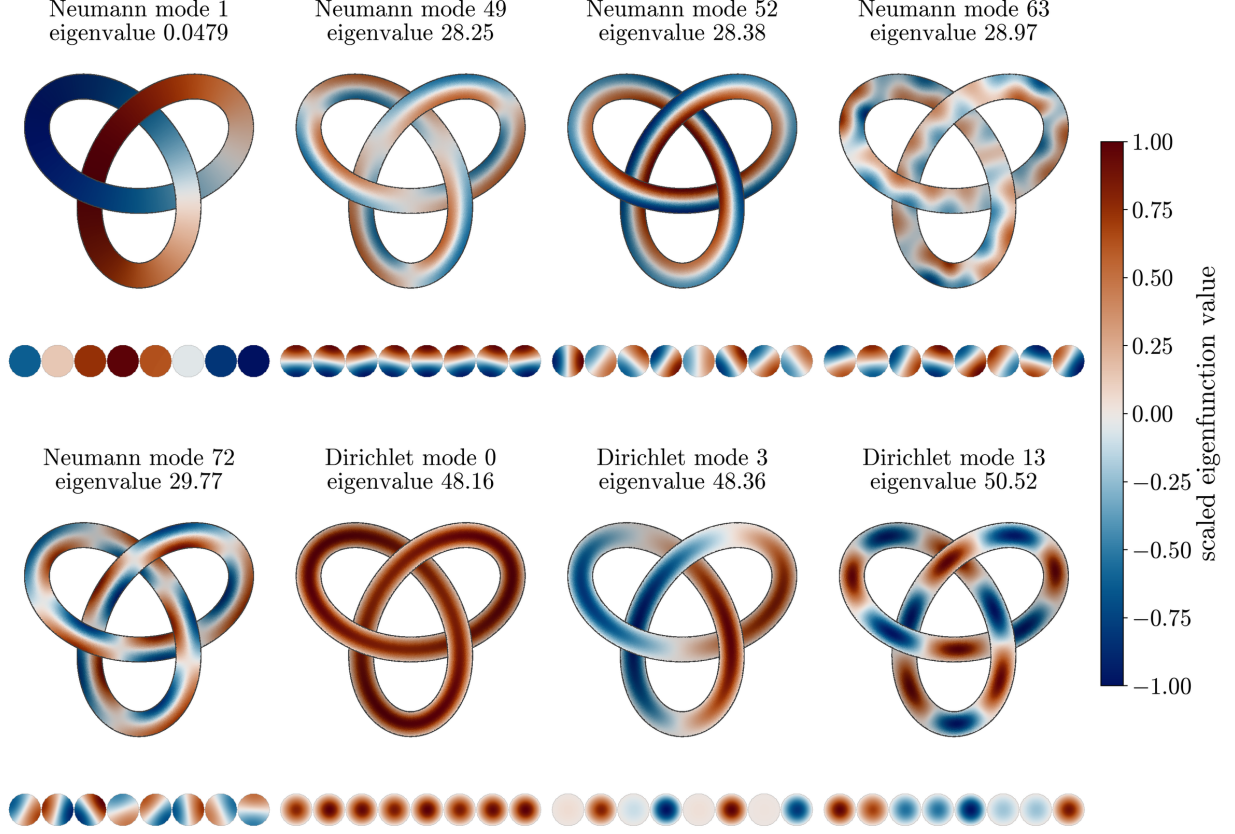


Figure 8: Scalar eigenfunctions of the solid trefoil tube with the metric (8.14), cross-section radius $a = 0.35$, and Neumann or Dirichlet conditions on the lateral boundary; $p = 3$, 78,048 unknowns. The ribbon is the longitudinal section through the centerline that faces the viewer. The disks are eight equally spaced cross-sections in the Frenet coordinates (u, v) .

N	p	DOFs	E_{L^2}	s_{L^2}	E_{H^1}	s_{H^1}
2	2	64	1.5133×10^{-1}	—	1.6405×10^0	—
3	2	216	9.2024×10^{-2}	1.23	1.2663×10^0	0.64
4	2	512	4.3183×10^{-2}	2.63	8.2533×10^{-1}	1.49
3	3	729	2.1148×10^{-2}	—	4.8483×10^{-1}	—
4	3	1728	7.4950×10^{-3}	3.61	2.4400×10^{-1}	2.39
6	3	5832	1.5060×10^{-3}	3.96	8.0059×10^{-2}	2.75
3	4	1728	4.9081×10^{-3}	—	1.5338×10^{-1}	—

Table 8: Anisotropic scalar refinement for $g_{ij} = e^{2\phi} B_{ij}$, with ϕ and B specified in (8.2)–(8.3). The $p = 3$ sequence includes the $N = 6$ mesh with 1296 tetrahedra; all seven runs use the full non-diagonal inverse metric. Orders compare successive available mesh sizes at fixed degree; the $p = 4$ entry has one mesh size.

r	k	local dimension	algebraic basis κ_2	Bernstein-QR basis κ_2
3	0	20	$3.0070 \cdot 10^3$	$4.0559 \cdot 10^1$
3	1	45	$1.6215 \cdot 10^4$	$2.5209 \cdot 10^2$
3	2	36	$4.7182 \cdot 10^4$	$7.3153 \cdot 10^2$
3	3	10	$1.3422 \cdot 10^4$	$3.5337 \cdot 10^2$
4	0	35	$2.0398 \cdot 10^5$	$5.0453 \cdot 10^2$
4	1	84	$1.6261 \cdot 10^6$	$4.3641 \cdot 10^3$
4	2	70	$6.5155 \cdot 10^6$	$1.6178 \cdot 10^4$
4	3	20	$1.8585 \cdot 10^6$	$7.4534 \cdot 10^3$

Table 9: Reference-element interpolation matrix condition numbers κ_2 on $\hat{K}$ in the stated coefficient norm. They measure the local basis against the moment degrees of freedom on the reference element.

cells	r	$\dim V_h^0$	$\dim V_h^1$	$\dim V_h^2$	$\dim V_h^3$	$\max D_{k+1} D_k $
5	1	1	6	10	5	$4.0 \cdot 10^{-15}$
5	2	7	32	45	20	$2.4 \cdot 10^{-14}$
5	3	23	93	120	50	$1.1 \cdot 10^{-13}$
40	2	54	254	360	160	
40	3	181	741	960	400	$6.4 \cdot 10^{-13}$
40	4	428	1628	2000	800	$4.8 \cdot 10^{-11}$

Table 10: Dimensions of global high-order constrained spaces on S^3/Γ with the round quotient metric (8.9) and the maximum absolute entry of the differential-composition matrix.

mesh	cells	variation of G_K in a cell	anisotropy
three-torus, Kuhn, $N = 6$	1296	$[0.751, 1.331]$	16.39
S^3/Γ	5	$[0.767, 0.995]$	4.00
Seifert–Weber	60	$[0.706, 5.228]$	20.29
Seifert–Weber	480	$[0.867, 1.612]$	60.73

Table 11: Measured chart constants. The variation tends to one under refinement. The refinement of the Seifert–Weber cells, which join the center of the dodecahedron to a face, increases the anisotropy.

8.5. Manufactured one-form problem

The experiment uses the periodic three-torus and the metrics (8.1)–(8.3). We prescribe an analytic periodic one-form and construct its source using the intrinsic codifferential.

On $M = \mathbb{R}^3/\mathbb{Z}^3$, use the metric (8.1) with ϕ as in (8.2) and $B = I_3$ or (8.3). Choose the smooth periodic one-form

$$\alpha = \cos(2\pi x_1) dx_2 + \sin(2\pi(x_1 + x_3)) dx_1. \quad (8.15)$$

It is not closed: the first summand has nonzero $dx_1 \wedge dx_2$ differential, and the second generally has a nonzero $dx_1 \wedge dx_3$ differential. Its codifferential is also nonzero for the stated nonconstant metric. Hence both terms of the one-form Hodge energy are nonzero.

Write $\alpha = \alpha_j dx_j$ and define $\beta = d\alpha$ with antisymmetric coefficients

$$\beta_{ij} = \partial_i \alpha_j - \partial_j \alpha_i. \quad (8.16)$$

Let g^{ij} be the entries of g^{-1} , g_{ij} those of g , and $\rho = \sqrt{\det g}$. The codifferential of the one-form is

$$\delta_g \alpha = -\rho^{-1} \partial_i (\rho g^{ij} \alpha_j). \quad (8.17)$$

For the two-form β , the coordinate coefficients of its codifferential can be computed by

$$(\delta_g \beta)_\ell = -\rho^{-1} g_{\ell j} \partial_i (\rho g^{ia} g^{jb} \beta_{ab}), \quad (8.18)$$

using Einstein summation. Equations (8.17)–(8.18) follow by integrating the L_g^2 inner product $(\beta, dv)_g$ by parts on the periodic manifold. They allow the exact source to be defined without introducing extrinsic vector calculus:

$$f = (d\delta_g + \delta_g d + I)\alpha. \quad (8.19)$$

All coefficients of f are analytic periodic functions of x . The identity term ensures that the source problem is uniquely solvable even in the presence of harmonic one-forms. The prescribed one-form α is the exact solution.

We solve the source problem (8.19) in the mixed formulation

$$(\sigma_h, \tau_h)_g - (\alpha_h, d\tau_h)_g = 0, \quad (d\sigma_h, v_h)_g + (d\alpha_h, dv_h)_g + (\alpha_h, v_h)_g = (f, v_h)_g, \quad (8.20)$$

for $(\sigma_h, \alpha_h) \in V_h^0 \times V_h^1$. The space $\mathcal{P}_r^- \Lambda^1$ contains all polynomial one-forms of degree at most $r - 1$, so for smooth fields and a stable complex the expected order at fixed r is $O(h^r)$ in the $H\Lambda^1$ graph norm and in L_g^2 [3–6]. The scalar experiments use the full space $\mathcal{P}_p \Lambda^0$, so errors at $p = r$ refer to different local spaces. We report the absolute errors $\|\alpha - \alpha_h\|_{L_g^2}$, $\|d(\alpha - \alpha_h)\|_{L_g^2}$, and $\|\delta_g \alpha - \sigma_h\|_{L_g^2}$ on meshes of $6N^3$ tetrahedra. On the finest pair of meshes at each order the observed L_g^2 orders are 0.99, 1.88, 2.94, and 3.80 for $r = 1, 2, 3, 4$, and the orders of the exterior derivative are 1.01, 1.97, 2.95, and 3.84, in agreement with $O(h^r)$. The auxiliary field σ_h converges faster, at the orders 1.85 and 2.95 for $r = 1, 2$.

N	r	$\dim V_h^1$	E_{L^2}	s_{L^2}	E_d	s_d	E_σ	s_σ
2	1	56	8.700×10^{-1}	–	4.141×10^0	–	4.476×10^0	–
3	1	189	6.725×10^{-1}	0.64	4.280×10^0	-0.08	4.030×10^0	0.26
4	1	448	5.694×10^{-1}	0.58	3.306×10^0	0.90	3.096×10^0	0.92
6	1	1512	4.214×10^{-1}	0.74	2.204×10^0	1.00	1.744×10^0	1.42
8	1	3584	3.232×10^{-1}	0.92	1.648×10^0	1.01	1.069×10^0	1.70
12	1	12096	2.161×10^{-1}	0.99	1.096×10^0	1.01	5.056×10^{-1}	1.85
2	2	304	4.467×10^{-1}	–	3.505×10^0	–	1.951×10^0	–
3	2	1026	2.634×10^{-1}	1.30	1.572×10^0	1.98	9.825×10^{-1}	1.69
4	2	2432	1.568×10^{-1}	1.80	9.472×10^{-1}	1.76	4.312×10^{-1}	2.86
6	2	8208	7.4532×10^{-2}	1.83	4.3810×10^{-1}	1.90	1.3287×10^{-1}	2.90
8	2	19456	4.3414×10^{-2}	1.88	2.4854×10^{-1}	1.97	5.6932×10^{-2}	2.95
2	3	888	2.435×10^{-1}	–	9.514×10^{-1}	–	1.084×10^0	–
3	3	2997	7.232×10^{-2}	2.99	4.648×10^{-1}	1.77	2.285×10^{-1}	3.84

Table 12: Manufactured one-form problem with $g_{ij} = e^{2\phi} \delta_{ij}$, using the same metric function as in (8.2). The errors are integrated with a separate quadrature rule. Rates use successive N values at fixed r according to (8.8).

The analytic source agrees with a centered-difference evaluation of the intrinsic codifferential to $3.1 \cdot 10^{-8}$ for $B = I_3$ and $2.6 \cdot 10^{-8}$ for (8.3). At $N = 2$, $r = 3$ the direct and Schur-complement solutions differ by $2.1 \cdot 10^{-14}$ in L_g^2 , and raising the assembly quadrature from order six to nine changes the L_g^2 error by $2.2 \cdot 10^{-5}$ against an error of $2.4345 \cdot 10^{-1}$. The relative algebraic residuals are below $3 \cdot 10^{-13}$ for the direct solves and at most $2.0 \cdot 10^{-11}$ for the Schur-complement solves, which use a conjugate gradient iteration preconditioned by a sparse factorization of $M_1 + D_1^T M_2 D_1$.

The higher-degree results are given in Table 14.

N	r	$\dim V_h^1$	E_{L^2}	s_{L^2}	E_d	s_d	E_σ
2	2	304	4.314×10^{-1}	—	3.306×10^0	—	1.737×10^0
3	2	1026	2.552×10^{-1}	1.29	1.480×10^0	1.98	8.879×10^{-1}
4	2	2432	1.523×10^{-1}	1.79	8.861×10^{-1}	1.78	3.987×10^{-1}

Table 13: Manufactured one-form problem with the non-diagonal metric $g_{ij} = e^{2\phi} B_{ij}$, where B is (8.3). Observed orders are calculated at fixed $r = 2$ as in (8.8).

N	r	$\dim V_h^1$	q	E_{L^2}	s_{L^2}	E_d	s_d	q_{err}
2	3	888	8	2.4345×10^{-1}	—	9.5146×10^{-1}	—	10
3	3	2997	6	7.2322×10^{-2}	2.99	4.6481×10^{-1}	1.77	8
4	3	7104	7	3.2634×10^{-2}	2.77	2.0593×10^{-1}	2.83	9
5	3	13875	7	1.7157×10^{-2}	2.88	1.0741×10^{-1}	2.92	9
6	3	23976	7	1.0045×10^{-2}	2.94	6.2722×10^{-2}	2.95	9
2	4	1952	8	4.4575×10^{-2}	—	5.7879×10^{-1}	—	10
3	4	6588	8	1.5732×10^{-2}	2.57	1.0663×10^{-1}	4.17	10
4	4	15616	8	5.2710×10^{-3}	3.80	3.5351×10^{-2}	3.84	10
2	5	3640	9	2.4844×10^{-2}	—	8.9102×10^{-2}	—	11

Table 14: Higher-order one-form calculations for $g_{ij} = e^{2\phi} \delta_{ij}$, with ϕ as in (8.2). Errors are absolute. The orders q and q_{err} are the one-dimensional Gauss orders in assembly and error integration, while s_{L^2} and s_d are observed mesh-refinement orders. The $r = 5$ row gives the error level at $N = 2$.

To separate local approximation from the mixed solve, we computed the L_g^2 projection of the exact one-form (8.15) onto the discontinuous space

$$W_{h,r}^1 = \prod_{K \in \Delta} \mathcal{P}_r^-(K), \quad E_{\text{br}} = \inf_{v \in W_{h,r}^1} \|\alpha - v\|_{L_g^2(M)}. \quad (8.21)$$

Each cell projection uses the exact variable metric (8.1) with $B = I_3$ and a separate Duffy integration rule of order twelve. Since $V_{h,r}^1 \subset W_{h,r}^1$, every conforming discrete solution satisfies

$$E_{\text{br}} \leq \inf_{v_h \in V_{h,r}^1} \|\alpha - v_h\|_{L_g^2(M)} \leq \|\alpha - \alpha_h\|_{L_g^2(M)}. \quad (8.22)$$

Table 15 reports the projected error and the corresponding mixed-method error, both in absolute intrinsic L_g^2 norm. The mixed-method errors exceed the broken best-approximation errors by factors

N	r	E_{br}	$\ \alpha - \alpha_h\ _{L_g^2}$	$\ \alpha - \alpha_h\ _{L_g^2}/E_{\text{br}}$
2	3	1.39844×10^{-1}	2.43454×10^{-1}	1.74
3	3	4.02273×10^{-2}	7.23220×10^{-2}	1.80
2	4	2.87556×10^{-2}	4.45750×10^{-2}	1.55
3	4	9.12061×10^{-3}	1.57320×10^{-2}	1.72

Table 15: Discontinuous best-approximation errors for the manufactured one-form with $g_{ij} = e^{2\phi} \delta_{ij}$ and (8.2), compared with the conforming mixed-method errors from Table 14. The broken projection is computed cellwise using the same trimmed local spaces and an independent quadrature rule; its value is a numerical lower bound, to integration accuracy, for the error attainable on the specified mesh and polynomial space.

between 1.55 and 1.80, so the errors in Table 14 are governed by local resolution. Along a body

diagonal of a Freudenthal tetrahedron the phase $2\pi(x_1 + x_3)$ in (8.15) changes by $4\pi/N$, which is a full period across one element at $N = 2$.

On the flat torus the constant one-form $\alpha = dx_1$ lies in $V_{h,r}^1$ and solves $(\Delta_{H,g} + I)\alpha = \alpha$. At $N = 2$ the computed L_g^2 errors are $3.32 \cdot 10^{-15}$, $1.37 \cdot 10^{-13}$, and $5.01 \cdot 10^{-12}$ for $r = 1, 2, 3$, with exterior-derivative errors of the same size. The one-form reference moment matrices have condition numbers $2.52 \cdot 10^2$, $4.36 \cdot 10^3$, and $9.47 \cdot 10^4$ for $r = 3, 4, 5$, and the maximum entries of $D_{k+1}D_k$ on the $N = 2$ mesh are $5.62 \cdot 10^{-13}$, $4.49 \cdot 10^{-11}$, and $3.97 \cdot 10^{-9}$. Both quantities grow with r by about one and two orders of magnitude per step, which sets the range $r \leq 5$ of the form computations reported here.

8.6. Effect of metric approximation

For a cell K with affine periodic coordinates $x = x_K + E_K \hat{x}$, set $x_{K,c} = x_K + E_K(1/4, 1/4, 1/4)^T$. We compare the exact reference metric (8.4) with the cellwise-constant metric

$$G_{K,h}^{\text{fr}} = e^{2\phi(x_{K,c})} E_K^T B E_K. \quad (8.23)$$

For each pair of calculations, the mesh, periodic coefficient identifications, polynomial degree, and integration rule are identical; only the element metric is changed. The sampled relative metric error is

$$\varepsilon_{g,h} = \max_K \max_{\hat{x}_q} \left| e^{2(\phi(x_{K,c}) - \phi(x_K + E_K \hat{x}_q))} - 1 \right|, \quad (8.24)$$

where $\hat{x}_q$ are the element quadrature points. Table 16 lists the first positive scalar eigenvalues of the two discrete pencils on the same finite element space. The quantity $\varepsilon_{g,h}$ is a maximum over quadrature points. For the comparison with Theorem 5.10 we therefore enclose ϕ on every coordinate cube by interval arithmetic, which gives $t_-(N) \leq \phi(x_{K,c}) - \phi(x) \leq t_+(N)$ on every tetrahedron and the ratio interval (5.9). The bounds in Table 17 are rounded outward. The frozen tensor (8.23) is a measurable cellwise tensor in the classification of Section 2: its tangential components jump across facets, so it lies outside the Regge class of [23], and Theorem 5.10 is the applicable bound.

N	p	DOFs	$\varepsilon_{g,h}$	$\mu_{h,1}(g)$	$\mu_{h,1}(g_h^{\text{fr}})$	$ \Delta\mu_{h,1} $
2	2	64	0.530	34.573394	33.277525	1.295870
3	2	216	0.495	34.633200	35.362260	0.729060
4	2	512	0.480	34.110626	34.784837	0.674211
6	2	1728	0.368	33.879958	34.172905	0.292947
8	2	4096	0.275	33.837301	34.007653	0.170352
3	3	729	0.513	33.851389	34.676863	0.825474
4	3	1728	0.485	33.823156	34.533445	0.710289

Table 16: Matched metric replacement on the three-torus with $B = I_3$. g_h^{fr} is defined in (8.23); the sampled relative metric discrepancy is (8.24). Both spectra use the same periodic Bernstein trial space.

The measured shifts are one to four percent, well inside the intervals; at $\varepsilon = 0.48$ in dimension three the upper factor of Theorem 5.9 is 5.473. The bounds correspond to the worst-case coefficient configuration. Barycenter freezing of a smooth conformal factor cancels the first-order term of the perturbation on each cell, so the observed shifts decrease like h^2 : they fall by factors 2.30 and 1.72 from $N = 4$ to 6 and from 6 to 8, against 2.25 and 1.78 for second order.

N	p	$t_-(N)$	$t_+(N)$	Predicted ratio interval	Observed ratio
2	2	-0.2200	0.2200	[0.414, 2.411]	0.962518
3	2	-0.3052	0.2652	[0.332, 3.257]	1.021051
4	2	-0.2182	0.2182	[0.417, 2.394]	1.019765
6	2	-0.1786	0.1786	[0.489, 2.043]	1.008647
8	2	-0.1391	0.1391	[0.573, 1.745]	1.005034
3	3	-0.3052	0.2652	[0.332, 3.257]	1.024385
4	3	-0.2182	0.2182	[0.417, 2.394]	1.021000

Table 17: Coefficient-level metric comparison for the matched isotropic three-torus calculations. The predicted intervals are obtained from analytical interval bounds on ϕ in each coordinate cube and (5.9), then rounded outward. The observed ratio is $\mu_{h,1}(g_h^{\text{fr}})/\mu_{h,1}(g)$. The bounds are uniform in the finite element degree but deliberately conservative.

At $N = 4$ the exact-metric eigenvalues are 34.110626 at $p = 2$ and 33.823156 at $p = 3$, and the frozen-metric values are 34.784837 and 34.533445, so the metric replacement is resolved separately from the polynomial approximation. The three $p = 2$ metric shifts at $N = 4, 6, 8$ have consecutive rates $\log(0.674211/0.292947)/\log(6/4) = 2.06$ and $\log(0.292947/0.170352)/\log(8/6) = 1.88$. For a smooth coefficient frozen at an affine element barycenter, the linear term in its Taylor expansion has zero mean over the cell. When the relevant eigenfunctions have uniformly controlled cellwise derivatives, this first-moment cancellation yields a second-order contribution to their Rayleigh quotients. The pointwise coefficient comparison in Theorem 5.10 remains an unconditional bound on the fixed discrete space.

N	p	$\mu_{h,1}(g)$	$\mu_{h,1}(g_h^{\text{fr}})$	$ \Delta\mu_{h,1} $
2	2	32.118045	31.137090	0.980955
3	2	32.206856	32.600857	0.394001
4	2	31.723236	32.119819	0.396583

Table 18: Matched metric replacement for $g_{ij} = e^{2\phi}B_{ij}$ with B given by (8.3); the mesh and scalar finite element space are unchanged within each row.

8.7. Hyperbolic quotient and geometric approximation

The Seifert–Weber hyperbolic dodecahedral space is treated by three computations. The scalar values in Table 19 and the piecewise-flat comparison come from a model-coordinate Bernstein implementation; the scalar eigenfunctions and the coexact one-form spectrum below are computed with the spaces $S_p^0(\Delta)$ and $V_{h,r}^k$ of this paper. Its reference-cell metric is the pullback of the hyperbolic metric in the hyperboloid model. With $\eta = \text{diag}(-1, 1, 1, 1)$ and an element map $F_K : \widehat{K} \rightarrow \mathbb{H}^3 \subset \mathbb{R}^4$, the intrinsic reference tensor is $G_K = (DF_K)^T \eta DF_K \in \mathbb{R}^{3 \times 3}$. The mesh is a geodesically refined dodecahedron with the prescribed face identifications, and the basis is the model-coordinate Bernstein basis of the accompanying code.

A paired geometric-replacement calculation holds the degree-two trial space fixed on a finer mesh of 30,720 cells and replaces the hyperbolic element metric by the piecewise-flat metric obtained from the model’s edge-length data. The saved output gives

	exact hyperbolic metric	piecewise-flat metric
vol_h	11.19906474	11.32032468
$\mu_{h,1}$	9.571131	9.498565

cells	degree	unknowns	first positive eigenvalue	first-cluster range
480	2	632	9.893001	[9.893001, 10.12309]
480	4	5104	9.571478	[9.571478, 9.572981]
480	6	17256	9.570139	[9.570139, 9.570144]

Table 19: Scalar eigenvalues on the Seifert–Weber hyperbolic manifold from the separate model-coordinate Bernstein implementation. The first-cluster range contains six computed values. The volume at degree six is 11.1990647408 in the unit-curvature normalization.

The volume discrepancy of the piecewise-flat metric is 0.12126. The piecewise-flat metric is determined by the edge lengths, so its tangential components are single-valued on every facet and it is a Regge metric; the source-problem analysis of [23] applies to it, and Theorem 5.9 bounds the eigenvalue shift. The barycenter freezing in Section 8.6 is of the measurable cellwise class.

A second computation uses the ordinary space $S_6^0(\Delta)$ of Section 3 in the chart $y = V_K b$, $x = y/r$ with $r^2 = -y^T \eta y$, whose pullback metric is

$$G_K = E_K^T \left(\frac{\eta}{r^2} + \frac{(\eta y)(\eta y)^T}{r^4} \right) E_K. \quad (8.25)$$

On the 480-cell mesh it has 17,256 unknowns, volume 11.1990647, and first level 9.5701 with six members, in agreement with Table 19. The next computed levels are 15.3637, 19.3280, 19.4929, 32.1693, 32.2051, 32.8616, 34.3539, 42.0991, 44.1142, and 45.7429, with multiplicities 6, 6, 4, 4, 6, 6, 5, 4, 6, 5. Every multiplicity is four, five, or six. The rotations of the dodecahedron commute with the face pairings and act on the quotient by isometries. Projection of each eigenspace onto the functions invariant under the rotations about a five-fold or a three-fold axis gives two invariant functions for each axis in a level of multiplicity six, none and two in a level of multiplicity four, and one and one in a level of multiplicity five. These counts agree with the restriction of three-, four- and five-dimensional representations of the icosahedral rotation group to its cyclic subgroups. Figure 9 shows eight of these functions.

Coexact one-forms. This computation tests the assembled form complex of Section 4 on a hyperbolic quotient against results from geometric topology. Let λ_1^* denote the first eigenvalue of the Hodge Laplacian on coexact one-forms. Lin and Lipnowski show that a hyperbolic rational homology sphere with $\lambda_1^* > 2$ is an L -space [32]. For the Seifert–Weber space they prove $\lambda_1^* > 2.0004717$, observe that their trace-formula function indicates $\lambda_1^* \in [2.03883, 2.04586]$ with a six-dimensional eigenspace, and prove that every eigenvalue $\lambda^* \leq 64$ has multiplicity at least four [31]. We solve

$$D_1^T M_2 D_1 u = \lambda M_1 u \quad (8.26)$$

with the trimmed spaces of order r , the facet pairings of the dodecahedron, and the metric (8.25), on the 60-cell mesh and on its refinement with 480 cells. The kernel of (8.26) is $\ker D_1$, which for this rational homology sphere equals $\operatorname{im} D_0$. An eigenvector with $\lambda > 0$ satisfies $\lambda D_0^T M_1 u = D_0^T D_1^T M_2 D_1 u = 0$, so it lies in $\mathcal{C}_h^1 = (\operatorname{im} D_0)^\perp_{M_1}$, the discrete coexact space, and the positive spectrum of (8.26) is the coexact spectrum; the pencil is solved as it stands. The eigenvalues are computed by Lanczos iteration with shift-invert at $\sigma = 3$, which lies between the first two levels, and the kernel appears as a cluster at 10^{-9} . On 60 cells at $r = 4$ the six eigenvectors have relative residuals below 10^{-9} , and the M_1 -norm of their component in $\operatorname{im} D_0$ is below $6 \cdot 10^{-11}$ of their norm.

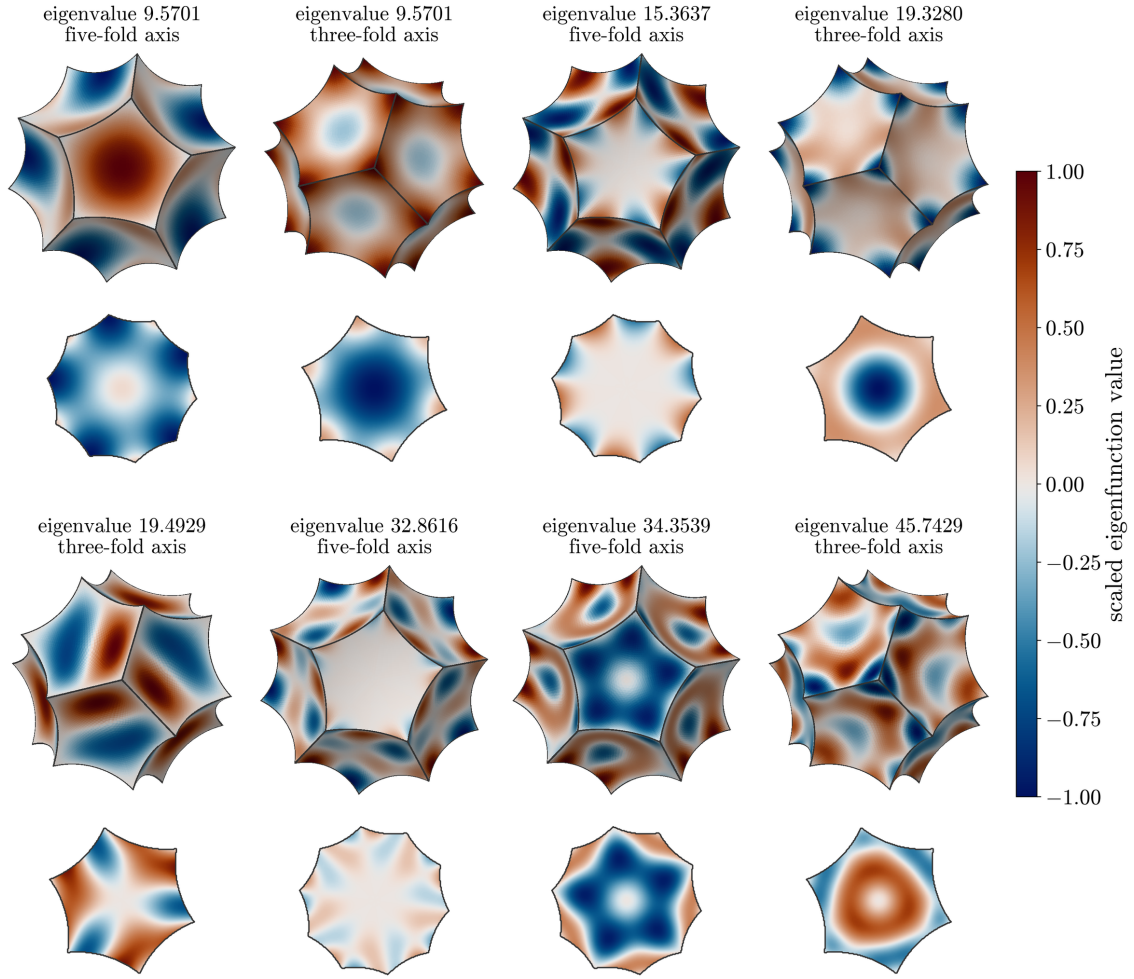


Figure 9: Eigenfunctions on the Seifert–Weber hyperbolic dodecahedral space, computed with $S_6^0(\Delta)$ on 480 cells. Each function is drawn on the boundary of the fundamental dodecahedron in the Poincaré ball model and, below it, on the plane section through the center orthogonal to the viewing direction. The faces are geodesic planes, which appear as spherical caps in this model. Each function is invariant under the rotations about the stated axis.

In the dense solve at $r = 4$ on 60 cells the kernel of (8.26) has dimension $631 = \dim V_{h,4}^0 - 1$, so the first Betti number of the assembled complex is zero, as for a rational homology sphere. Table 20 lists the six lowest positive eigenvalues. The continuum eigenvalue is sixfold, so the spread of the six computed values measures the discretization error. On 480 cells it decreases from $1.4 \cdot 10^{-3}$ at $r = 2$ to $2 \cdot 10^{-6}$ at $r = 4$, where the six values lie in $[2.039273, 2.039275]$; the second level is 5.48919 with five members. Raising the Gauss order from 8 to 11 at $r = 4$ on 60 cells leaves six digits unchanged.

A dense solve on the 60-cell mesh at $r = 4$ gives 203 positive eigenvalues below 66; Weyl’s law predicts about 194 below 64. They form 52 clusters, all of sizes three, four, five, or six. The 60-cell mesh is invariant under the rotation group of the dodecahedron, so the clusters are irreducible representations of that group, and the triples occur in close pairs that form the six-dimensional representations of the full isometry group. The computed values agree with the interval and with the

cells	r	$\dim V_{h,r}^1$	smallest of the six	largest of the six
60	3	1104	2.045336	2.051499
60	4	2432	2.040767	2.040998
60	5	4540	2.039523	2.039724
480	2	3032	2.038071	2.039513
480	3	8868	2.039325	2.039478
480	4	19504	2.039273	2.039275

Table 20: The six lowest positive eigenvalues of (8.26) on the Seifert–Weber space, computed with the trimmed complex of order r and Gauss–Duffy order $r + 4$. The interval indicated in [31] is $[2.03883, 2.04586]$.

multiplicity statement of [31]. The computed values converge under the hypotheses of Theorem 5.8. The discrete coexact space $\mathcal{C}_h^1$ differs from the continuous one, so the approach to the limit is two-sided: the $r = 2$ values on 480 cells lie below it and the $r \geq 3$ values above.

9. Conclusion

The elementwise pullback metric and the facet trace maps are the data from which high-order scalar and compatible differential-form finite elements are assembled on a Riemannian manifold. The scalar space is a constrained Bernstein–Bézier space, and the form spaces are assembled from canonical polynomial traces. The stiffness and Hodge matrices carry the metric; the exterior derivative and its cochain identities are fixed by the reference spaces and the mesh identifications. For $n = 2$ and a metric conformal to the round sphere the construction returns the weighted weak forms of [27].

The scalar estimates follow from conformity with the prescribed metric and uniform local interpolation, and the form estimates use the bounded commuting-projection framework [4, 5, 34], which the averaging construction extends to finite free isometric quotients. The metric perturbation bounds order every positive discrete Hodge eigenvalue when the local tensor is replaced on a fixed finite element space. They hold for Regge metrics, where they complement the source-problem analysis of [23], and for cellwise tensors with jumps.

On the trefoil tube surface and on the variable-metric three-torus the scalar errors converge at the orders $p + 1$ and p , through degree six on the surface, and eigenvalues through index 480 converge on meshes refined along the knot. For the one-form problem the errors converge at the order h^r for $r = 1, \dots, 4$, the mixed-method errors stay within a factor 1.8 of the discontinuous best approximation, and the constant-form test reproduces the solution to 10^{-11} . On the Poincaré dodecahedral space, degree fourteen on five cells gives the level 168 with absolute error $3 \cdot 10^{-11}$ and the multiplicities 13, 21, 25, 31, and 33 of the next levels. On the Seifert–Weber space the trimmed complex of order four on 480 cells gives the lowest coexact one-form eigenvalue as 2.03927 with six members, and the clusters below 64 have sizes three to six, in agreement with the trace-formula interval and the multiplicity theorem of Lin and Lipnowski [31]. The solid-tube computations give scalar Dirichlet and Neumann spectra on polygonal cross sections, including the rotation of the transverse Neumann modes with the torsion of the centerline. The eigenfunction figures display the symmetry, multiplicity, and localization of the computed modes on every manifold.

CRediT authorship contribution statement

Shelvean Kapita: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The programs, meshes, facet pairings, run scripts, and the data behind every table and figure of Sections 7 and 8 are deposited at [\[repository name and DOI, to be inserted by the author before submission\]](#). The deposit states the conventions of the metric interface $\hat{x} \mapsto G_K(\hat{x})$ and of the facet permutations, and it lists for every table the mesh, the order, the Gauss order, the eigensolver, and its tolerance.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude Fable 5.1 (Anthropic) for programming support, for checking computations and derivations, for literature search, and for drafting and editing parts of the text. The author conceived the method, directed and checked the computations, reviewed and edited the content, and takes full responsibility for the content of the published article.

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